

## Chapter 2      Fuzzy Sets

### Key Concepts

Classical / Fuzzy sets, Fuzziness / Vagueness / Inexactness, Membership functions, Singleton, Fuzzy membership, Normality, Height, Support, Core, Fuzzy membership function, Triangular function, Trapezoidal function, Gaussian function, S-function, Normalization, Dilation, Concentration, Contrast Intensification, Fuzzification, Fuzzy relations, Relation matrix, Composition, Max-min composition, Fuzzy Cartesian product, Level set,  $\alpha$ -cut,  $\alpha$ -cut decomposition theorem, Restricted scalar multiplication, Fuzzy cardinality, Fuzzy extension principle

### Chapter Outline

- 2.1    *Crisp Sets : A Review*
- 2.2    *Fuzzy Sets*
- 2.3    *Fuzzy Membership Functions*
- 2.4    *Operations on Fuzzy Sets*
- 2.5    *Fuzzy Relations*
- 2.6    *Fuzzy Extension Principle*

- Chapter Summary
- Bibliography & Historical Notes
- Solved Problems
- Test Your Knowledge
- Exercise

This chapter presents the basic concepts of fuzzy set theory. The fuzzy set theory is an extension, or generalization, of the classical, occasionally referred to as crisp, set theory in the sense that the later can be considered as a special case of the former. Hence the chapter starts with a review of the fundamentals of the classical set theory. Elementary concepts of fuzzy set theory, e.g., membership functions, transformations on membership functions, linguistic variables, fuzzy set operations etc. are presented along with illustrative examples in sections 2.2, 2.3 and 2.4. This is followed by a discussion on fuzzy relations and related matters in section 2.4. The chapter ends with a presentation of the fuzzy extension principle that provides a technique to map a function from the crisp domain to its equivalent in the fuzzy domain.

## 2.1 Crisp Sets : A Review

The notion of a set is of fundamental importance in Mathematics. It can be informally defined in the following way.

**Definition 2.1 ( Set )** A **set** is a collection of distinct *elements*, or *members*, without repetition and without ordering. ♣

**Example 2.1 ( Set )** The set atomic particles = { *electron, proton, neutron* }. Since a set is defined by its members without regard to ordering or repetitions, the set of atomic particles cited above is identical to the set { *proton, neutron, electron* }, or the set { *neutron, proton, neutron, electron, proton, proton* }. ♣

There are two ways to describe a set. The most obvious one is by enumerating the members of the set. However, occasionally it is convenient to describe a set by citing a property common to all members of the set. Both of these two kinds of set notations are exemplified below.

**Example 2.2 ( Set Notations )** The sets *A, B, C, D* given below are described by enumeration of its members.

$A = \{ \text{Jack, Jill, hill, pail, water} \}$

$B = \{ +, -, \times, \div \}$

$C = \{ \text{Socrates, toothbrush, loneliness, 235} \}$

$D = \{ \text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday} \}$

The set *D* of all days of a week can be equivalently expressed by citing the property shared by its member. Thus,

$D = \{ x \mid x \text{ is a day of the week} \}$

A few other sets described in similar way are

$E = \{ x \mid x \text{ is an integer, and } -1 \leq x \leq +1 \} = \{ -1, 0, +1 \}$

$F = \{ x \mid x \text{ is a prime number} \}$

$G = \{ x \mid x \text{ is a polygon} \}$

$H = \{ x \mid x \text{ is an element having 8 electrons in its outermost shell} \}$  ♣

The elementary concepts of classical set theory are presented below.

### 2.1.1 Basic Concepts

This subsection provides the basic set theoretic concepts e.g. cardinality of a set, the null set, the universal set, belongingness to a set, subset, superset and so on.

**Definition 2.2** ( *Cardinality of a Set* ) The number of elements in a set  $S$  is termed as its **cardinality** and is denoted as  $|S|$ . ♣

**Example 2.3** ( *Cardinality of a set* ) A set can be finite or infinite, depending on whether it has a finite or infinite number of elements. For example, the sets  $A, B, C, D$  and  $H$  in Example 2.2 are finite sets with cardinalities  $|A| = 5, |B| = 4, |C| = 4$ , and  $|D| = 7$ . What is the size of  $H$ ? Among the sets mentioned in Example 2.2 the sets  $F$  and  $G$  are infinite sets. ♣

**The Null Set and the Universal Set** There are two specially interesting sets, viz. the **null ( empty )** set and the **universal** set. These are usually represented by the symbols  $\emptyset$  and  $U$  respectively. The null set is the set without any member, so that  $|\emptyset| = 0$ . The universal set, on the other hand, is the set of all possible elements in a certain context.

**Example 2.4** ( *The null set and the universal set* ) Let  $X$  be the set of all natural numbers that are divisible by 4 but not divisible by 2. As there is no such integer which is divisible by 4 but not divisible by 2,  $X = \emptyset$ . Similarly, the set of positively charged electrons is also the null set. Moreover, we may define a set  $S = \{x \mid x \text{ is blue-eyed}\}$  in the context of the universal set of all human beings. Then  $S$  is the set of all blue-eyed persons and  $U$  is the set of all human beings. However, if  $U$  is the set of all living creatures, then  $S$  is the set of all blue-eyed creatures including, but not limited to, all blue-eyed human beings. ♣

An element  $x$  is said to **belong** to a set  $S$ , expressed as  $x \in S$ , if  $x$  is included in  $S$ . Otherwise  $x$  does not belong to  $S$ , written as  $x \notin S$ . For example, for the set  $F = \{x \mid x \text{ is a prime number}\}$ ,  $3 \in F$ , as 3 is a prime number, but  $4 \notin F$ .

**Definition 2.3** ( *Subset* ) A set  $T$  is said to be a **subset** of set  $S$ , written as  $T \subseteq S$ , if every element of  $T$  is in  $S$ , i.e.,  $\forall x$  if  $x \in T$ , then  $x \in S$ . Equivalently,  $S$  is **superset** of  $T$ , symbolized as  $S \supseteq T$ , if and only if  $T$  is a subset of  $S$ . ♣

$T$  is a **proper subset** of  $S$ , denoted as  $T \subset S$ , if  $T \subseteq S$  and  $T \neq S$ . Hence, if  $T \subset S$ , then there is at least one member  $x$  such that  $x \in S$  but  $x \notin T$ . For an arbitrary set  $S$  the following properties are obvious from the definitions cited above.

- i)  $\emptyset \subseteq S$
- ii)  $S \subseteq U$
- iii)  $S \subseteq S$

Moreover, the chain rule applies to set inclusion operation, i.e., for any three sets  $A, B, C$ , if  $A \subseteq B$  and  $B \subseteq C$ , then  $A \subseteq C$ .

**Definition 2.4 ( Equality of Sets )** Two sets  $S$  and  $T$  are **equal** if every element of  $S$  is in  $T$  and vice-versa. In other words,  $S = T$ , if and only if,  $S \subseteq T$  and  $T \subseteq S$ . ♣

**Definition 2.5 ( Power Set )** Given a set  $S$ , the **power set** of  $S$ , denoted as  $P ( S )$ , or  $2^S$ , is the set of all subsets of  $S$ . ♣

**Example 2.5 ( Power Set )** Let us consider the set  $S = \{ \text{black}, \text{white} \}$ . Then  $P ( S ) = 2^S = \{ \emptyset, \{ \text{black} \}, \{ \text{white} \}, \{ \text{black}, \text{white} \} \}$ . Obviously, if  $| S | = n$ , then  $| P ( S ) | = | 2^S | = 2^n$ . Similarly, if  $S = \{ \text{electron}, \text{proton}, \text{neutron} \}$ , then  $P ( S ) = 2^S = \{ \emptyset, \{ \text{electron} \}, \{ \text{proton} \}, \{ \text{neutron} \}, \{ \text{electron}, \text{proton} \}, \{ \text{proton}, \text{neutron} \}, \{ \text{electron}, \text{neutron} \}, \{ \text{electron}, \text{proton}, \text{neutron} \} \}$ . ♣

### 2.1.2 Operations on Sets

There are three basic operations on sets, viz. union (  $\cup$  ), intersection (  $\cap$  ), and complementation (  $'$  ).

**Definition 2.6 ( Set Union )** Given two sets  $P$  and  $Q$ , the **union** of  $P$  and  $Q$ , denoted as  $P \cup Q$ , is the set of all elements either in  $P$ , or in  $Q$ , or in both  $P$  and  $Q$ .

$$P \cup Q = \{ x \mid x \in P \text{ or } x \in Q \} \quad \clubsuit$$

**Definition 2.7 ( Set Intersection )** Given two sets  $P$  and  $Q$ , the intersection of  $P$  and  $Q$ , denoted as  $P \cap Q$ , is the set of all elements both in  $P$ , and  $Q$ .

$$P \cap Q = \{ x \mid x \in P \text{ and } x \in Q \} \quad \clubsuit$$

**Definition 2.8 ( Complement of a Set )** The complement of  $P$ , denoted as  $P'$ ,  $\overline{P}$ ,  $\neg P$ , or  $\sim P$ , is the set of all elements ( of  $U$  ) outside  $P$ .

$$P' = \{ x \mid x \notin P \} \quad \clubsuit$$

There are various notations for complementation, as indicated in the definition. However, in this book we shall denote the complement of  $P$  as  $P'$ .

**Definition 2.9 ( Difference Between Sets )** The difference of set  $P$  from  $Q$ , denoted as  $P - Q$ , is the set of all elements in  $P$  but not in  $Q$ .

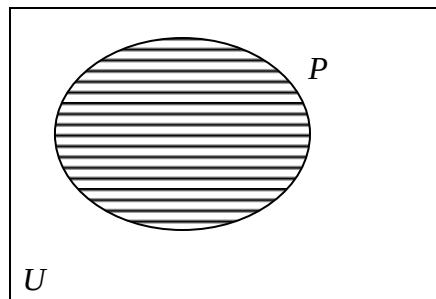
$$P - Q = \{ x \mid x \in P \text{ and } x \notin Q \} \quad \clubsuit$$

It is easy to prove that  $P - Q = P \cap Q'$

**Definition 2.10** ( *Symmetric Difference* )    The symmetric difference of  $P$  and  $Q$ , denoted as  $P \oplus Q$ , is the set of all elements that are either in  $P$ , or in  $Q$ , but not in both  $P$  and  $Q$ .

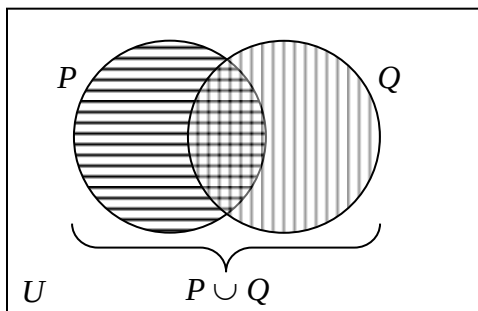
$$\begin{aligned} P \oplus Q &= \{ x \mid (x \in P \text{ and } x \notin Q), \text{ or } (x \in Q \text{ and } x \notin P) \} \\ &= (P - Q) \cup (Q - P) \\ &= (P \cap Q') \cup (P' \cap Q) \end{aligned}$$

♣

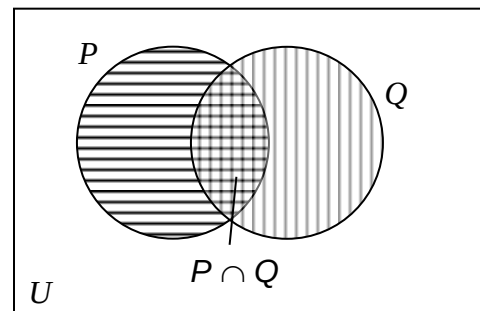


**Fig. 2.1.** Venn diagram of set  $P$

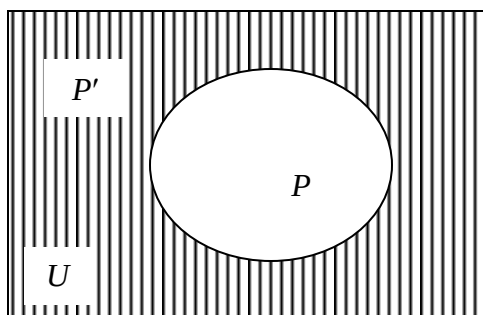
**Venn Diagrams**    Quite often it is convenient to represent a set theoretic expression visually with the help of a diagram called Venn diagrams. Usually, a Venn Diagram consists of a rectangle presenting the universal set  $U$  with other sets presented with the help of circles / ovals inside the rectangle. For example, the Venn diagram of a set  $S$  is presented in Fig. 2.1. The region inside the oval is the set  $S$ . The Venn diagrams for union, intersection and complementation are shown in Fig. 2.2, 2.3 and 2.4 respectively. Fig. 2.5 depicts the difference of two sets while Fig. 2.6 presents the operation of symmetric difference.



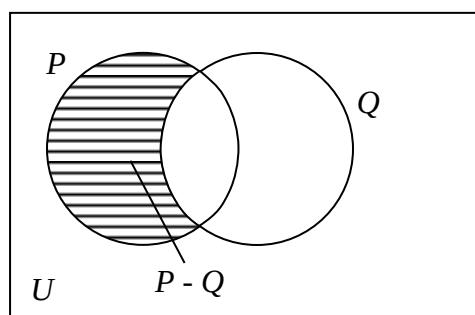
**Fig. 2.2.** Union



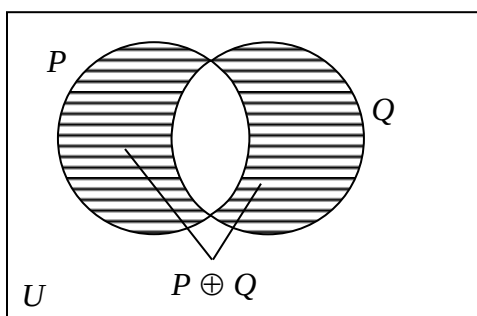
**Fig. 2.3.** Intersection



**Fig. 2.4.** Compliment of a set



**Fig. 2.5.** Set difference



**Fig. 2.6.** Symmetric difference

**Definition 2.11** ( *Cartesian Product* ) Let  $P$  and  $Q$  be two sets. The Cartesian product of  $P$  and  $Q$ , denoted as  $P \times Q$ , is the set of all ordered pairs  $(x, y)$  such that  $x \in P$  and  $y \in Q$ .

$$P \times Q = \{ (x, y) \mid x \in P \text{ and } y \in Q \}$$



### 2.1.3 Properties of Sets

Certain properties are obeyed by the set theoretic operations of union, intersection, complementation, symmetric difference etc. These properties are summarized below.

|             |                           |
|-------------|---------------------------|
| Idempotency | $A \cup A = A$            |
|             | $A \cap A = A$            |
| Commutative | $A \cup B = B \cup A$     |
|             | $A \cap B = B \cap A$     |
|             | $A \oplus B = B \oplus A$ |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Associative            | $A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) = (A \cap B) \cup C$ $A \oplus (B \oplus C) = (A \oplus B) \oplus C$                                                                                                                                                                                                                                                                                                               |
| Distributive           | $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ <p>( Left distributivity of union over intersection )</p> $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ <p>( Right distributivity of intersection over union )</p> $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ <p>( Left distributivity of intersection over union )</p> $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$ <p>( Right distributivity of union over intersection )</p> |
| De Morgan's law        | $(A \cap B)' = A' \cup B'$ $(A \cup B)' = A' \cap B'$                                                                                                                                                                                                                                                                                                                                                                                         |
| Identity               | $A \cup \emptyset = \emptyset \cup A = A$ $A \cap \emptyset = \emptyset \cap A = \emptyset$ $A \cup U = U \cup A = U$ $A \cap U = U \cap A = A$                                                                                                                                                                                                                                                                                               |
| Involution             | $(A')' = A$                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Law of excluded middle | $A \cup A' = U$                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Law of contradiction   | $A \cap A' = \emptyset$                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Universal compliment   | $U' = \emptyset$ $\emptyset' = U$                                                                                                                                                                                                                                                                                                                                                                                                             |
| Absorption             | $A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$                                                                                                                                                                                                                                                                                                                                                                                               |

A set theoretic identity of the form L.H.S = R.H.S may be proved by showing that L.H.S  $\subseteq$  R.H.S and simultaneously L.H.S  $\supseteq$  R.H.S. Example 2.6 illustrates the technique by proving the absorption properties. Also, the properties mentioned above may be applied to prove set theoretic identities. A few such examples are given in the section Solved Problems.



**Example 2.6 ( Proof of Absorption Properties )** The absorption properties state that for arbitrary sets  $A$  and  $B$ , the identities  $A \cup ( A \cap B ) = A$  and  $A \cap ( A \cup B ) = A$ . These can be proved as follows.

Let us take an arbitrary element  $x$  of  $A$ . Now

|                                              |                                  |
|----------------------------------------------|----------------------------------|
| Let $x \in A$                                | : Assumption                     |
| $\therefore x \in A \cup B$                  | : By definition of union         |
| $\therefore x \in A \cap ( A \cup B )$       | : By definition of intersection  |
| $\therefore A \subseteq A \cap ( A \cup B )$ | : By definition of set inclusion |

|                                              |                                  |
|----------------------------------------------|----------------------------------|
| Now, Let $x \in A \cap ( A \cup B )$         | : Assumption                     |
| $\therefore x \in A$ and $( A \cup B )$      | : By definition of intersection  |
| $\therefore x \in A$                         |                                  |
| $\therefore A \cap ( A \cup B ) \subseteq A$ | : By definition of set inclusion |

Hence  $A \cap ( A \cup B ) = A$

|                                                               |                               |
|---------------------------------------------------------------|-------------------------------|
| Again, $A \cup ( A \cap B ) = ( A \cup A ) \cap ( A \cup B )$ | : Distributive law            |
| $= A \cap ( A \cup B )$                                       | : $A \cup A = A$              |
| $= A$                                                         | : $A \cap ( A \cup B ) = A$ ♣ |

## 2.2 Fuzzy Sets

Fuzzy set theory is a generalization of the classical set theory. Unlike the later, fuzzy set theory recognizes and incorporates in its formalism the natural vagueness that we the human beings are habituated to deal with in our practical, daily, life. This section presents the fundamental concepts of the fuzzy set theory.

### 2.2.1 Fuzziness / Vagueness / Inexactness

It is quite normal to utter sentences like ‘He is a *rich* man’, or ‘The car is *very expensive*’, or ‘*Old* people tend to be *weak*’ and so on in our everyday conversation. While such expressions as *rich*, *very*, *expensive*, *old*, *weak* etc. are extremely convenient in practical communication they are characteristically inexact in the sense that there are no well-defined demarcations between *rich* and *poor*, *very* and *little*, *expensive* and *cheap*, *old* and *young*, or *weak* and *strong*. A person can be rich as well as poor simultaneously, of course to different extents. This vagueness is desirable because otherwise everyday conversation would have been impossible. However, the classical set theory is not equipped to handle such vagueness as it does not allow an element to be a partial member, and also a partially non-member, of a set simultaneously. Therefore classical set theory is inadequate to model our intuitive notion of a *set* in general.

As stated earlier, Fuzzy set theory is a generalization of the classical set theory so that the classical set theory is a special case of the fuzzy set theory. It takes into consideration the natural vagueness that we the human beings deal with in our practical, daily, life. As an example, let us consider the data related to a family as described in TABLE 2.1.

**TABLE 2.1. A FAMILY DATA SET**

| # | Family member   | Age | Gender |
|---|-----------------|-----|--------|
| 1 | <i>Grand-pa</i> | 72  | Male   |
| 2 | <i>Grand-ma</i> | 63  | Female |
| 3 | <i>Dad</i>      | 41  | Male   |
| 4 | <i>Mom</i>      | 38  | Female |
| 5 | <i>Daughter</i> | 15  | Female |
| 6 | <i>Son</i>      | 13  | Male   |
| 7 | <i>Aunty</i>    | 52  | Female |

It is customary to refer to a classical set as *crisp* in order to differentiate it from a *fuzzy* set. The crisp set of the family members  $U = \{ \textit{Grand-pa}, \textit{Grand-ma}, \textit{Dad}, \textit{Mom}, \textit{Sister}, \textit{Brother}, \textit{Aunt} \}$  may be treated as the *reference set*, or the *universe of discourse*. Now, consider the sets  $M$  and  $F$  of the male family members and female family members respectively. These are crisp sets because for any arbitrary element  $x$  of  $U$ , it is possible to decide precisely whether  $x$  is member of the set, or not. There is no intermediate status regarding the membership of  $x$  to the respective set. However, deciding the membership of an arbitrary  $x \in U$  to the set of *senior* persons of the family is not as straightforward as in case of  $M$  or  $F$ . The status of membership of an element  $x$  with respect to a given set  $S$  is expressed with the help of a *membership function*  $\mu$ . A set, crisp or fuzzy, may be defined in terms of membership function.

## 2.2.2 Set Membership

Description of a set in terms of its membership function is presented in this subsection. We first define crisp membership which is followed by the fuzzy membership function.

**Definition 2.12 ( Membership Function )** Given an element  $x$  and a set  $S$ , the **membership** of  $x$  with respect to  $S$ , denoted as  $\mu_S(x)$ , is defined as

$$\mu_S(x) = \begin{cases} 1, & \text{if } x \in S \\ 0, & \text{if } x \notin S \end{cases} \quad \clubsuit$$

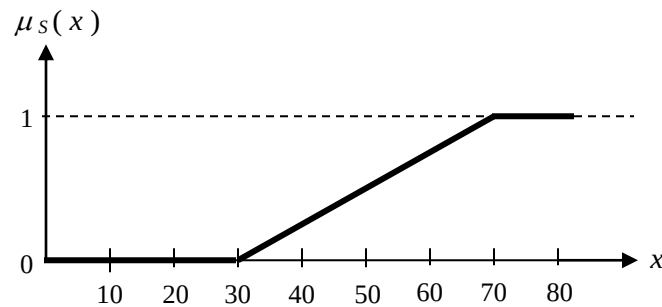
**Example 2.7 ( Set Membership )** Let us consider the set  $M$  of male family members and set  $F$  of female family members with reference to the family presented in TABLE 2.1. We see that  $\mu_M( Dad ) = 1$ , and  $\mu_M( Mom ) = 0$ . Similarly,  $\mu_F( Dad ) = 0$ , and  $\mu_F( Mom ) = 1$ . Membership values of the other family members in  $M$  and  $F$  can be ascertained in similar manner. ♣

Now, consider  $A$  to be the set of *senior* persons in the family. *Seniority* is a familiar and frequently-used attribute to a person. But is there any clear and unambiguous way to decide whether a person should be categorized as senior or not? Let us see with reference to  $U$ , the universe of discourse.

We may agree without any hesitation that Grand-pa, being 72 years old, is a senior person, so that  $Grand-pa \in A$ . On the other hand the brother and the sister are both too young to be categorized as senior persons. Therefore, we may readily accept that  $Daughter \notin A$  and  $Son \notin A$ . What about *Mom*, or *Dad*? They are not as aged as Grand-pa but neither as young as the daughter or the son. Moreover, Grand-ma is almost a senior person, being at 63 years, but she might be categorized as a middle-aged person as well.

The point is, the concept of a senior person is not as clearly defined as the gender of the person. In fact, there is a whole range of gray area between total inclusion and total exclusion, over which the degree of membership of a person in the set of senior persons varies.

This intuitive notion of partial membership of an element in a set can be formalized if one allows the membership function to assume any real value between 0 and 1, including both. This means that an element may belong to a set to any extent within the range  $[ 0, 1 ]$ . Hence it is now possible for an element  $x$  to be 0.5 member, or  $1/\sqrt{2}$  member of a set  $S$  so that we may say  $\mu_S( x ) = 0.5$ , or  $\mu_S( x ) = 1/\sqrt{2}$ .



**Fig. 2.7.** Membership function for the fuzzy set of senior family members

**Membership profiles / functions** Quite often it is convenient to express the membership of various elements with respect to a fuzzy set with the help of a function, referred to as the **membership function**. Take for example the fuzzy set  $A$  of senior persons on the universe of discourse  $U$  described in TABLE 2.1. For any  $x \in U$ , we may

determine the membership value of  $x$  in  $A$  with the help of the following membership function.

$$\mu_A(x) = \begin{cases} 0, & \text{if } x < 30 \\ \frac{x-30}{40}, & \text{if } 30 \leq x < 70 \\ 1, & \text{if } x \geq 70 \end{cases} \quad (2.1)$$

Here  $x$  represents the age of the concerned person. The nature of the membership function is shown graphically in Fig. 2.7.

Formula 2.1 may be applied to find to what extent a person is a member of the fuzzy set  $A$  of senior persons. For example, as Grand-pa is more than 70 years old, he is a full member of  $A$ ,  $\mu_A(\text{Grand-pa}) = 1.0$ . However for Grand-ma we have  $\mu_A(\text{Grand-ma}) = (63 - 30) / 40 = 33 / 40 = 0.825$ . Hence Grand-ma is a senior person to a large extent, but not as fully as Grand-pa. TABLE 2.2 shows the membership values of all the family members.

**TABLE 2.2. MEMBERSHIP VALUES OF THE SENIOR FAMILY MEMBERS**

| # | Family member   | Age | $\mu_A(x)$ |
|---|-----------------|-----|------------|
| 1 | <i>Grand-pa</i> | 72  | 1.0        |
| 2 | <i>Grand-ma</i> | 63  | 0.825      |
| 3 | <i>Dad</i>      | 41  | 0.275      |
| 4 | <i>Mom</i>      | 38  | 0.200      |
| 5 | <i>Daughter</i> | 15  | 0.0        |
| 6 | <i>Son</i>      | 13  | 0.0        |
| 7 | <i>Aunty</i>    | 52  | 0.55       |

If we want to describe such a fuzzy set by explicitly enumerating its members we need to indicate the membership value of each element along with the element itself. Therefore, the fuzzy set  $A$  of senior persons can be expressed as a set of ordered pairs  $A = \{ (\text{Grand-pa}, 1.0), (\text{Grand-ma}, 0.825), (\text{Dad}, 0.275), (\text{Mom}, 0.2), (\text{Aunty}, 0.55) \}$ . *Daughter* and *Son* are not in the list because it is customary not to mention the members with zero membership value. Thus fuzzy sets can be formally defined as given below.

### 2.2.3 Fuzzy Sets

This subsection presents the definition of a fuzzy set along with a few illustrative examples.

**Definition 2.13 ( Fuzzy set )** A **fuzzy set**  $F$  on a given universe of discourse  $U$  is defined as a collection of ordered pairs  $(x, \mu_F(x))$  where  $x \in U$ , and for all  $x \in U$ ,  $0.0 \leq \mu_F(x) \leq 1.0$ .

$$F = \{ (x, \mu_F(x)) \mid x \in U, 0.0 \leq \mu_F(x) \leq 1.0 \} \quad \clubsuit$$

Classical sets, often referred to as *crisp* sets to distinguish them from fuzzy sets, are special cases of fuzzy sets where the membership values are restricted to either 0, or 1. Each pair  $(x, \mu_F(x))$  of the fuzzy set  $F$  is known as a **singleton**.

**Notation ( Fuzzy sets )** Apart from enumerating the singletons as described above, fuzzy sets are frequently expressed as the union of all singletons where a singleton is denoted as  $\mu_F(x) / x$ . Using this notation

$$F = \sum_{x \in U} \mu_F(x) / x \quad (2.2)$$

Here the summation sign  $\sum$  is to be interpreted as union over all singletons, and not arithmetic sum. Formula 2.2 is appropriate for discrete sets. For continuous sets, the summation notation is replaced by the integral sign  $\int$ , as shown below.

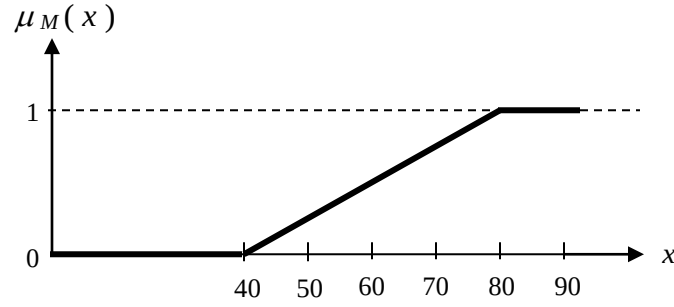
$$F = \int_U \frac{\mu_F(x)}{x} \quad (2.3)$$

**Example 2.8 ( Fuzzy Membership )** Let  $a, b, c, d$ , and  $e$  be five students who scored 55, 35, 60, 85 and 75 out of 100 respectively in Mathematics. The students constitute the universe of discourse  $U = \{a, b, c, d, e\}$  and a fuzzy set  $M$  of the students who are *good in Mathematics* is defined on  $U$  with the help of the following membership function.

$$\mu_M(x) = \begin{cases} 0, & \text{if } x < 40 \\ \frac{x-40}{40}, & \text{if } 40 \leq x < 80 \\ 1, & \text{if } x \geq 80 \end{cases} \quad (2.4)$$

The membership function is graphically shown in Fig. 2.8. Computing the membership value of each student with the help of the Formula 2.4 we get  $M = \{ (a, 0.375), (c, 0.5), (d, 1.0), (e, 0.875) \}$ , or equivalently

$$M = \frac{0.375}{a} + \frac{0.5}{c} + \frac{1.0}{d} + \frac{0.875}{e} \quad \clubsuit$$



**Fig. 2.8.** Membership function for students *good in Mathematics*

**Example 2.9 ( Fuzzy Membership )** Let us consider the infinite set of all real numbers between 0 and 1, both inclusive, to be the universe of discourse, or, the reference set,  $U = [ 0, 1 ]$ . We define a fuzzy set  $C0.5$  as the set of all real numbers in  $U$  that are *close* to 0.5 in the following way

$$C0.5 = \{ x \in [ 0, 1 ] \mid x \text{ is } \textit{close} \text{ to } 0.5 \}$$

The highest membership value would be attained by the point  $x = 0.5$  because that is the number closest to 0.5, and the membership is understandably 1. On the other hand since both 0 and 1 are furthest points from 0.5 ( within the interval  $[ 0, 1 ]$  ) they should have zero membership to  $C0.5$ . Membership values should increase progressively as we approach 0.5 from both ends of the interval  $[ 0, 1 ]$ . The membership function for  $C0.5$  may be defined in the following manner.

$$\mu_{C0.5}(x) = 1 - |2x - 1|, \quad \forall x \in [ 0, 1 ] \quad (2.5)$$

Since  $C0.5$  is a fuzzy set in a continuous domain it can be expressed with the help of the notation

$$C0.5 = \int_{x \in [0,1]} \frac{\mu_{C0.5}(x)}{x} = \int_{x \in [0,1]} \frac{1 - |2x - 1|}{x} \quad (2.6)$$

♣

## 2.2.4 Fuzzyness Vs. Probability

It must be appreciated that the membership of a fuzzy set relate to *vagueness* and not to probability which is a measure of *uncertainty*. The difference between fuzziness and probability is nicely explained with the help the following story. A man is wandering

for long hours in a desert and he is now dying for a glass of water. Someone offered him two glasses, glass A and glass B, of water with the condition that he can not drink from both the glasses and he has to choose only one of the two for drinking. The man has been informed that the water of glass A is poisonous to a degree of 0.1, which is not fatal. On the other hand glass B contains water which is poisonous with a probability of 0.1. Which glass of water should the thirsty man choose to quench his thirst, and of course, remain alive? If he drinks from glass B he has 90% chance of having pure drinking water but there is 10% chance that the water may be poisonous and in that case he would certainly die. On the other hand if he drinks from glass A, he would fall ill due to the fact that the water is partially poisonous. However there is no chance of him dying because the degree of poisoning is not fatal. Glass A symbolizes fuzziness while glass B symbolizes probability.

### 2.2.5 Features of Fuzzy Sets

Fuzzy sets are often characterized with certain features, *e.g.*, *normality*, *height*, *support*, *core*, *cardinality* etc. These features reveal the nature and structure of a fuzzy set. These features are briefly explained in this subsection.

**Definition 2.14 ( Normality )** A fuzzy set  $F$  is said to be **normal** if there exists an element  $x$  that completely belongs to  $F$ ,  $\mu_F(x) = 1$ . A fuzzy set which is not normal is said to be **sub-normal**. ♣

**Definition 2.15 ( Height )** The height of a fuzzy set is defined as the maximal membership value attained by its elements. ♣

$$height(F) = \max_{x \in U} \mu_F(x) \quad (2.7)$$

$U$  is the universe of discourse, or, the reference set, for  $F$ , and  $height(F)$  is the height of the fuzzy set  $F$ . Obviously,  $F$  is a normal fuzzy set if  $height(F) = 1$ .

**Definition 2.16 ( Support )** The **support** of a fuzzy set  $F$ , denoted by  $supp(F)$ , is the set of all elements of the reference set  $U$  with non-zero membership to  $F$ . ♣

$$Supp(F) = \{ x \mid x \in U, \& \mu_F(x) > 0 \} \quad (2.8)$$

**Definition 2.17 ( Core )** The *core* of a fuzzy set  $F$  is the set of all elements of the reference set  $U$  with complete membership to  $F$ . ♣

$$Core(F) = \{ x \mid x \in U, \& \mu_F(x) = 1 \} \quad (2.9)$$

Both  $supp(F)$  and  $core(F)$  are crisp sets. It is easy to see that  $core(F) \subseteq supp(F)$ .

**Definition 2.18 ( Cardinality )** The sum of all membership values of the members of a fuzzy set  $F$  is said to be the **cardinality** of  $F$ . ♣

$$|F| = \sum_{x \in U} \mu_F(x) \quad (2.10)$$

**Example 2.10 ( Cardinality )** Let us consider the fuzzy set  $M$  defined on the reference set  $U = \{ a, b, c, d, e \}$  as described in Example 2.8.

$$M = \frac{0.375}{a} + \frac{0.5}{c} + \frac{1.0}{d} + \frac{0.875}{e}$$

The fuzzy set  $M$  is normal, because we have  $\mu_M(d) = 1.0$ . Its height is 1.0. Moreover, as we see,  $\text{supp}(M) = \{ a, c, d, e \}$ ,  $\text{core}(M) = \{ d \}$ , and the cardinality of  $M$  is  $|M| = 0.375 + 0.5 + 1.0 + 0.875 = 2.750$ . ♣

## 2.3 Fuzzy Membership Functions

Theoretically any function  $\mu_F : U \rightarrow [0, 1]$  may act as the membership function for a fuzzy set  $F$ . The nature of the membership function depends on the context of the application. For example, what is hot with respect to human body temperature is certainly very cool for a blast furnace. Hence the membership function for the concept *hot* must differ in the two contexts mentioned above. Fuzzy sets are usually described with simple membership functions. A few parameterized functions that are quite frequently used for this purpose are given below along with their graphical representations.

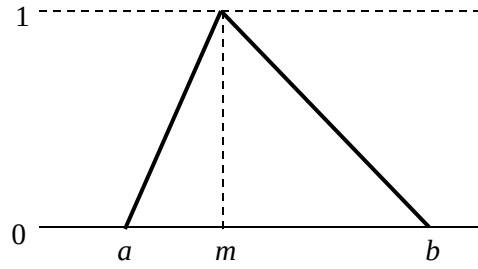
### 2.3.1 Some Popular Fuzzy Membership Functions

This subsection presents the Triangular function, Trapezoidal function, Gaussian function and S-function as the widely used fuzzy membership functions.

#### a) The Triangular Function ( See Eqn. 2.11, Fig. 2.9 )

$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{m-a}, & \text{if } a \leq x \leq m \\ \frac{b-x}{b-m}, & \text{if } m \leq x \leq b \\ 0, & \text{if } x \geq b \end{cases} \quad (2.11)$$

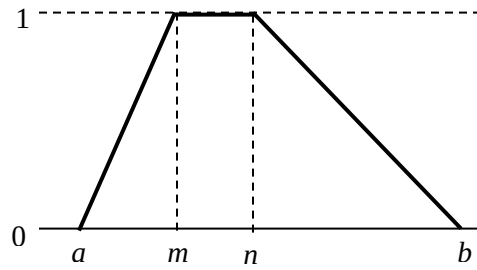




**Fig. 2.9.** Shape of Triangular function

**b) Trapezoidal Function** ( See Eqn. 2.12 and Fig. 2.10 )

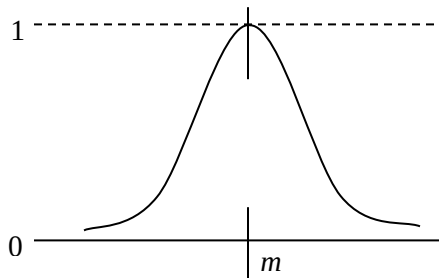
$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{m-a}, & \text{if } a \leq x \leq m \\ 1, & \text{if } m \leq x \leq n \\ \frac{b-x}{b-n}, & \text{if } n \leq x \leq b \\ 0, & \text{if } x \geq b \end{cases} \quad (2.12)$$



**Fig. 2.10.** Trapezoidal function

**c) Gaussian Function** ( See Eqn. 2.13 and Fig. 2.11 )

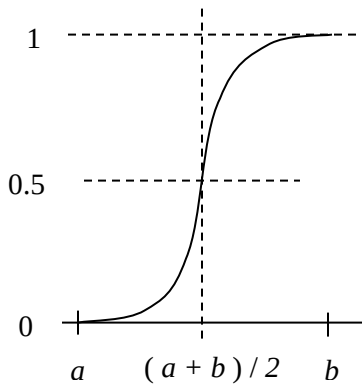
$$\mu(x) = e^{-k(x-m)^2} \quad \text{where } k > 0 \quad (2.13)$$



**Fig. 2.11.** Gaussian function

c) **S-function** ( See Eqn. 2.14 and Fig. 2.12 )

$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ 2\left(\frac{x-a}{b-a}\right)^2, & \text{if } a \leq x \leq m \\ 1 - 2\left(\frac{x-b}{b-a}\right)^2, & \text{if } m \leq x \leq b \\ 1, & \text{if } x \geq b \end{cases} \quad (2.14)$$



**Fig. 2.12.** The S - function

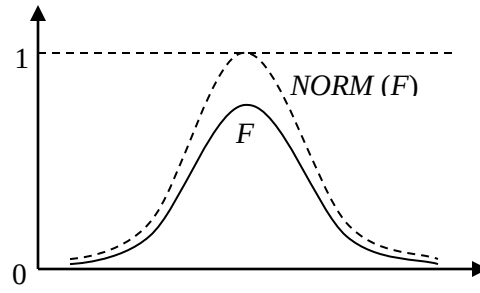
The point  $m = (a + b) / 2$  is known as the crossover point of the S-function.

### 2.3.2 Transformations

There are a few simple but useful transformations often applied to fuzzy membership functions. The so called *normalization*, *dilation*, *concentration*, *contrast intensification* and *fuzzification* are widely used. These transformations are briefly discussed in this subsection. The effects of these transformations on Gaussian functions are shown graphically.

**a) Normalization** The normalization operation converts a subnormal fuzzy set  $F$  to a normal fuzzy set. It is obtained by dividing each membership value by the height of the fuzzy set ( See Eqn. 2.15 and Fig. 2.13 ).

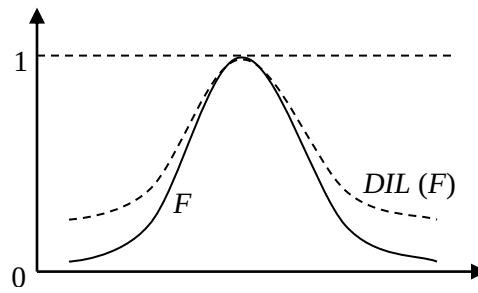
$$NORM ( F, x ) = \frac{\mu_F ( x )}{height( F )} \quad (2.15)$$



**Fig. 2.13.** Normalization

**b) Dilation** This operation ‘flattens’ the membership function so that it attains relatively higher values and consequently the overall shape of the membership function gets dilated. In other words, the resultant function is less pointed around the higher membership values. The effect is shown in Fig. 2.14 with respect to the Gaussian function.

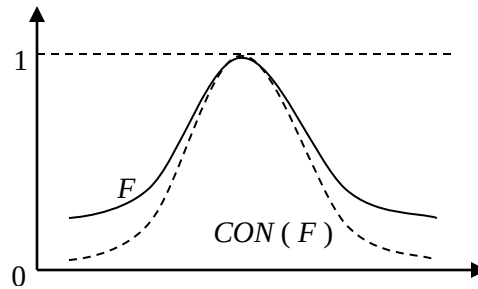
$$DIL(F, x) = [\mu_F(x)]^{1/2} \quad (2.16)$$



**Fig. 2.14.** Dilation

**c) Concentration** Concentration has the reverse effect of dilation. Here the function attains relatively lower values and consequently it gets more pointed around the higher membership values. The formula for concentration is given by Eqn. 2.17 and Fig. 2.15 shows the effect on Gaussian function.

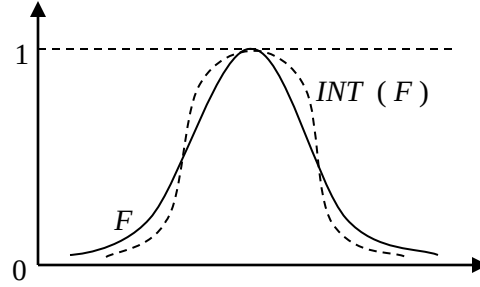
$$CON(F, x) = [\mu_F(x)]^2 \quad (2.17)$$



**Fig. 2.15.** Concentration

**d) Contrast Intensification** Contrast intensification is achieved by reducing the membership values that are less than 0.5 and elevating those with values higher than 0.5 ( see Eqn. 2.18 and Fig. 2.16 ).

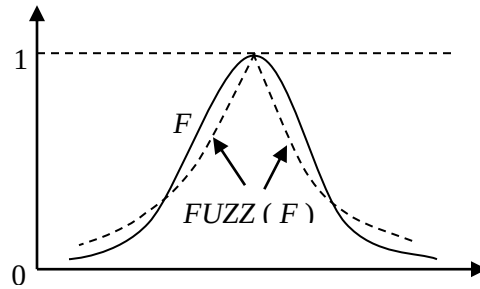
$$INF (F, x) = \begin{cases} 2[\mu_F(x)]^2, & \text{if } 0 \leq \mu_F(x) \leq 0.5 \\ 1 - 2[1 - \mu_F(x)]^2, & \text{otherwise} \end{cases} \quad (2.18)$$



**Fig. 2.16.** Contrast intensification

**e) Fuzzification** Fuzzification produces the reverse effect of contrast intensification. Here the membership values that are less than 0.5 are elevated while those with a value more than 0.5 are reduced. The mathematical formula for fuzzification is given in Eqn. 2.19 and its effect on Gaussian function is shown in Fig. 2.17.

$$FUZZ (F, x) = \begin{cases} [\mu_F(x)/2]^{1/2}, & \text{if } 0 \leq \mu_F(x) \leq 0.5 \\ 1 - \sqrt{(1 - \mu_F(x))/2}, & \text{otherwise} \end{cases} \quad (2.19)$$

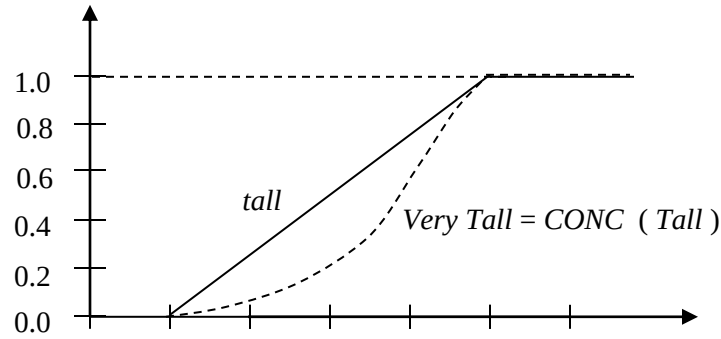


**Fig. 2.17.** Fuzzification

### 2.3.3 Linguistic Variables

The transformations presented above are useful while dealing with the so called **linguistic variables**. Conventional variables usually have numeric, or alphanumeric

string values. In contrast, a linguistic variable may have one of a number of allowable words, or phrases, as its value. Each of these legitimate linguistic values corresponds to a fuzzy set. For instance, consider the variable *Height*. As a conventional variable it may have any real number as its value. But as a linguistic variable its value is allowed to be one of a few predefined words, say  $\{ \textit{short}, \textit{average}, \textit{tall} \}$ . We may define a fuzzy set for each of the words *short*, *average*, and *tall*. Now, consider the attribute *very tall*. Here the word *very* may be viewed as a transformation applied to the fuzzy meaning of *tall*. One can reasonably implement *very* with the help of the concentration operation. The reason is *concentration* reduces the magnitude of the membership value. What is *tall* to some extent might be thought of *very tall* to a lesser extent. Therefore, the membership function for *very tall* should always be less than that of *tall* and moreover, the transition from low *very tall* to high *very tall* should be steeper than the transition from low *tall* to high *tall*. Fig. 2.18 gives an idea about the relationship described above.

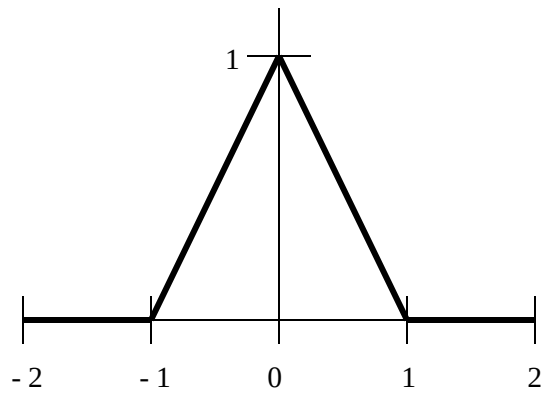


**Fig. 2.18.** Transformation on linguistic variable fuzzy set

**Example 2.11** ( *Transformations on fuzzy membership functions* ) Consider the fuzzy set  $F$  with a triangular membership function defined on the interval  $[-2, +2]$ .

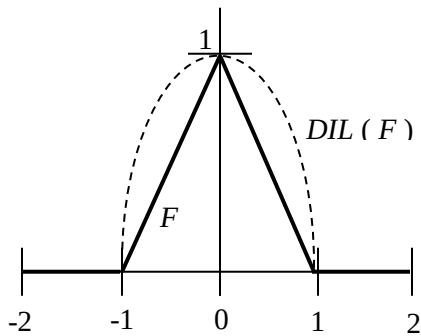
$$\mu_F(x) = \begin{cases} 0, & \text{if } -2 \leq x \leq -1 \\ x+1, & \text{if } -1 \leq x \leq 0 \\ 1-x, & \text{if } 0 \leq x \leq 1 \\ 0, & \text{if } 1 \leq x \leq 2 \end{cases} \quad (2.20)$$

Fig. 2.19 presents the shape of the triangular function defined above. Now let us apply the transformations normalization, dilation, concentration, contrast intensification and fuzzification on this membership function and see the resultant functional forms.

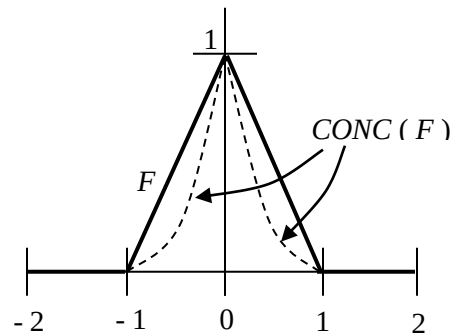


**Fig. 2.19.** Triangular membership function given by Eqn. 2.20

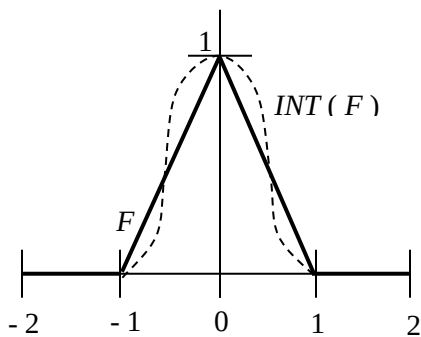
Since the height of the given membership function is 1, normalization has no effect on it. The forms of the resultant memberships due to the other operations, viz., dilation, concentration, contrast intensification, and fuzzification are shown in Fig. 2.20 - Fig. 2.23.



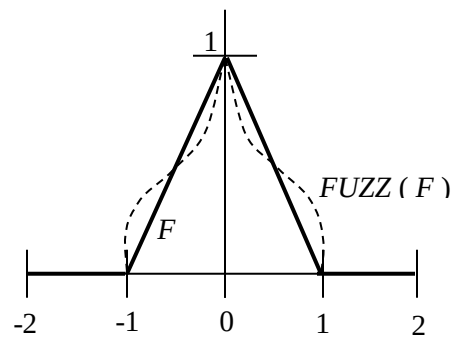
**Fig. 2.20.** Dilation



**Fig. 2.21.** Concentration



**Fig. 2.22.** Contrast intensification



**Fig. 2.23.** Fuzzification

## 2.4 Operations on Fuzzy Sets

Most of the usual set theoretic operations e.g., union, intersection, complementation etc. are readily extended to fuzzy sets. A list of such operators along with their definitions is shown in TABLE 2.3.

**TABLE 2.3.** OPERATIONS ON FUZZY SETS

| # | Operation/Relation               | Description                                                                                                                                   |
|---|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Union ( $P \cup Q$ )             | $\mu_{P \cup Q}(x) = \max \{ \mu_P(x), \mu_Q(x) \}, \forall x \in U$                                                                          |
| 2 | Intersection ( $P \cap Q$ )      | $\mu_{P \cap Q}(x) = \min \{ \mu_P(x), \mu_Q(x) \}, \forall x \in U$                                                                          |
| 3 | Complementation ( $P'$ )         | $\mu_{P'}(x) = 1 - \mu_P(x), \forall x \in U$                                                                                                 |
| 4 | Equality ( $P = Q$ )             | Two fuzzy sets $P$ and $Q$ are equal if and only if $\forall x \in U, \mu_P(x) = \mu_Q(x)$                                                    |
| 5 | Inclusion ( $P \subseteq Q$ )    | $P$ is included in $Q$ , i.e., $P$ is a subset of $Q$ , written as $P \subseteq Q$ , if and only if $\forall x \in U, \mu_P(x) \leq \mu_Q(x)$ |
| 6 | Product ( $P \cdot Q$ )          | $\mu_{P \cdot Q}(x) = \mu_P(x) \times \mu_Q(x), \forall x \in U$                                                                              |
| 7 | Difference ( $P - Q$ )           | $P - Q = P \cap Q'$                                                                                                                           |
| 8 | Disjunctive sum ( $P \oplus Q$ ) | $P \oplus Q = (P \cap Q') \cup (P' \cap Q)$                                                                                                   |

Obviously the union of two fuzzy sets is the *smallest* fuzzy set containing both and their intersection is the *largest* fuzzy set contained by both the given sets.

**Example 2.12 ( Fuzzy set operations )** Let us recall the reference set of the family members presented in TABLE 2.1 and the fuzzy set  $A$  of senior persons which is repeated here  $A = \{ ( Grand-pa, 1.0 ), ( Grand-ma, 0.825 ), ( Dad, 0.275 ), ( Mom, 0.2 ), ( Aunt, 0.55 ) \}$ , or, in the other notation

$$A = \frac{1.0}{Grand - pa} + \frac{0.825}{Grand - ma} + \frac{0.275}{Dad} + \frac{0.2}{Mom} + \frac{0.55}{Aunt}$$

The membership function for  $A$  is as follows

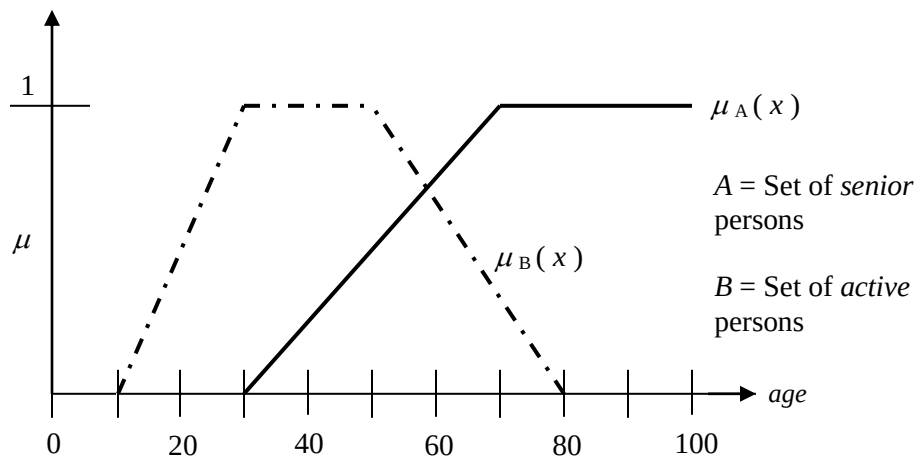
$$\mu_A(x) = \begin{cases} 0, & \text{if } \text{age}(x) \leq 30 \\ \frac{x-30}{40}, & \text{if } 30 \leq \text{age}(x) \leq 70 \\ 1, & \text{if } \text{age}(x) \geq 70 \end{cases} \quad (2.21)$$

We now introduce another fuzzy set  $B$  of *active* persons on the same reference set. The degree of activeness is assumed to be a function of the concerned person's age. Here is the membership function.

$$\mu_B(x) = \begin{cases} 0, & \text{if } \text{age}(x) \leq 10 \\ \frac{x-10}{15}, & \text{if } 10 \leq \text{age}(x) \leq 25 \\ 1, & \text{if } 25 \leq \text{age}(x) \leq 50 \\ \frac{80-x}{30}, & \text{if } 50 \leq \text{age}(x) \leq 80 \\ 0, & \text{if } \text{age}(x) \geq 80 \end{cases} \quad (2.22)$$

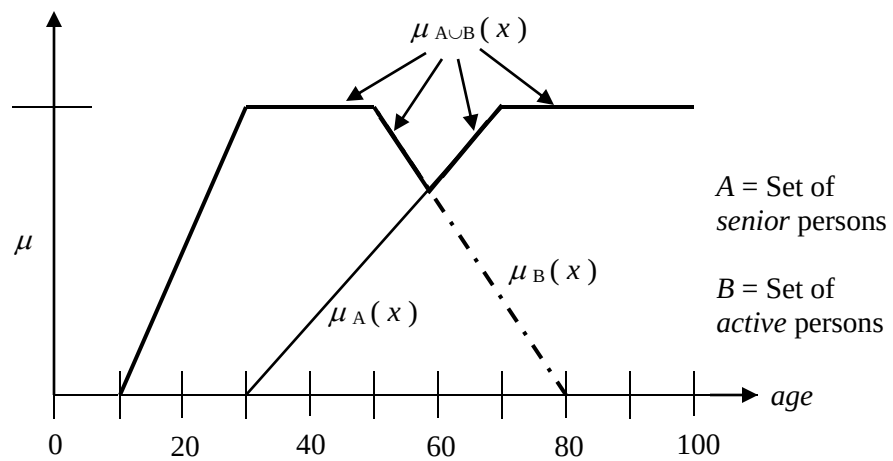
Fig. 2.24 presents the graphical view of the two membership functions. Computing the values of  $\mu_B(x)$  for each  $x$  in  $U$  using the membership function defined in formula 36.9 we get  $B = \{ (\text{Grand-pa}, 0.267), (\text{Grand-ma}, 0.567), (\text{Dad}, 1.0), (\text{Mom}, 1.0), (\text{Daughter}, 0.333), (\text{Son}, 0.2), (\text{Aunty}, 0.933) \}$ , or using the summation notation,

$$B = \frac{0.267}{\text{Grand-pa}} + \frac{0.567}{\text{Grand-ma}} + \frac{1.0}{\text{Dad}} + \frac{1.0}{\text{Mom}} + \frac{0.333}{\text{Daughter}} + \frac{0.2}{\text{Son}} + \frac{0.933}{\text{Aunty}}$$

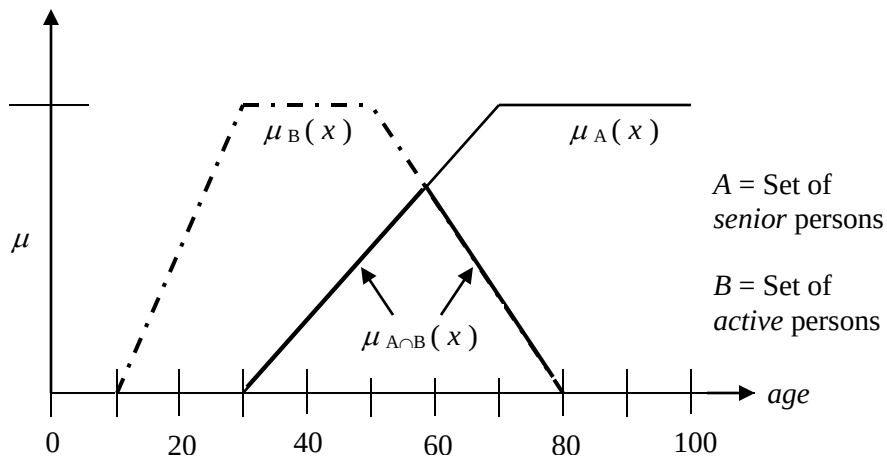


**Fig. 2.24.** Membership profiles of *aged* persons and *active* persons

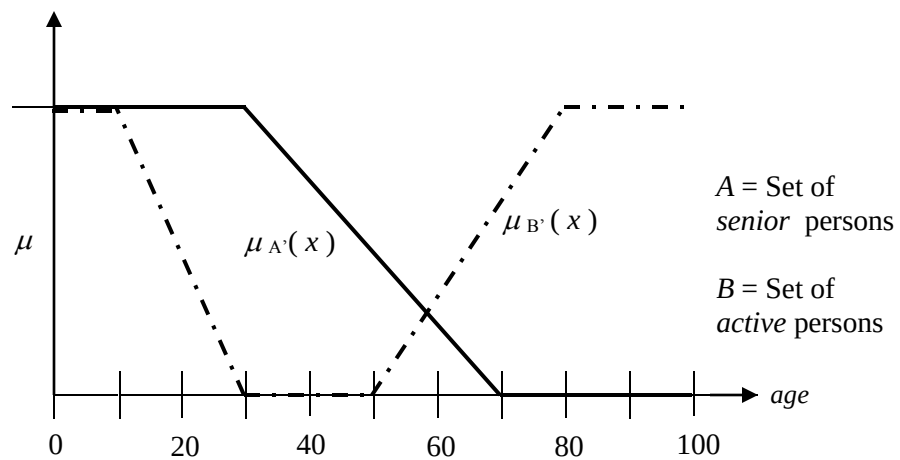




**Fig. 2.25.** Membership profile of *senior OR active* persons



**Fig. 2.26.** Membership profiles of *senior AND active* persons



**Fig. 2.27.** Membership profiles of NOT *senior* persons and NOT *active* persons

TABLE 2.4 shows the fuzzy sets  $A \cup B$ ,  $A \cap B$ ,  $A'$ , and  $B'$  obtained using definitions listed in TABLE 2.3. The profiles of the membership functions of the resultant fuzzy sets are depicted in Fig. 2.25, Fig. 2.26, and Fig. 2.27. ♣

**TABLE 2.4. OPERATIONS ON FUZZY SETS SENIOR & ACTIVE PERSONS**

| #  | Operation  |                           | Resultant fuzzy set                                                                                                                                                                                              |
|----|------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | $A \cup B$ | senior<br>OR<br>active    | $\{ (Grand-pa, 1.0), (Grand-ma, 0.825), (Dad, 1.0), (Mom, 1.0), (Daughter, 0.333), (Son, 0.2), (Aunty, 0.933) \}$                                                                                                |
| 2. | $A \cap B$ | senior<br>AND<br>Active   | $\{ (Grand-pa, 0.267), (Grand-ma, 0.567), (Dad, 0.275), (Mom, 0.2), (Aunty, 0.55) \}$                                                                                                                            |
| 3. | $A', B'$   | NOT senior,<br>NOT active | $A' = \{ (Grand-ma, 0.175), (Dad, 0.725), (Mom, 0.8), (Sister, 1.0), (Brother, 1.0), (Aunty, 0.45) \}$<br><br>$B' = \{ (Grand-pa, 0.733), (Grand-ma, 0.433), (Sister, 0.667), (Brother, 0.8), (Aunty, 0.067) \}$ |

**Properties of fuzzy set operations** Most of the properties of crisp sets, with the exception of a few, hold good in the area of fuzzy sets also. It is possible to verify on the basis of the definitions of various fuzzy set operations that the laws listed in TABLE 2.5 are satisfied by arbitrary fuzzy sets  $P$ ,  $Q$ , and  $R$  on some universe of discourse  $U$ . However, the relations  $P \cup P' = U$ , and  $P \cap P' = \Phi$  obeyed by crisp sets are no longer valid for fuzzy sets. For an arbitrary fuzzy set  $P$ , we have in general  $P \cup P' \neq U$ , and  $P \cap P' \neq \Phi$  because  $\mu_P(x) \in [0, 1]$ , and  $\mu_{P \cup P'}(x) = \max \{ \mu_P(x), 1 - \mu_P(x) \} \neq 1$ , and  $\mu_{P \cap P'}(x) = \min \{ \mu_P(x), 1 - \mu_P(x) \} \neq 0$ , unless either  $\mu_P(x) = 0$ , or  $\mu_P(x) = 1$ .

## 2.5 Fuzzy Relations

We have treated fuzzy sets as generalization of crisp sets where the degree of inclusiveness of an element may be anything from 0 to 1, and not just 0, or 1 as it is in case of crisp sets. In a similar fashion, the concept of a relation between two, or more, sets can be generalized to fuzzy relation. This section provides a review of the fundamentals of crisp relations, followed by a discussion on fuzzy relations.

**TABLE 2.5. FUZZY SET IDENTITIES**

| # | Law                 | Description                                                                                                  |
|---|---------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | Associativity       | (a) $(P \cup Q) \cup R = P \cup (Q \cup R)$<br>(b) $(P \cap Q) \cap R = P \cap (Q \cap R)$                   |
| 2 | Commutativity       | (a) $P \cup Q = Q \cup P$<br>(b) $P \cap Q = Q \cap P$                                                       |
| 3 | Distributivity      | (a) $P \cup (Q \cap R) = (P \cup Q) \cap (P \cup R)$<br>(b) $P \cap (Q \cup R) = (P \cap Q) \cup (P \cap R)$ |
| 4 | Idempotency         | (a) $P \cup P = P$<br>(b) $P \cap P = P$                                                                     |
| 5 | De Morgan's law     | (a) $(P \cup Q)' = P' \cap Q'$<br>(b) $(P \cap Q)' = P' \cup Q'$                                             |
| 6 | Boundary Conditions | (a) $P \cup \Phi = P, P \cup U = U$<br>(b) $P \cap \Phi = \Phi, P \cap U = P$                                |
| 7 | Involution          | $(P')' = P$                                                                                                  |
| 8 | Transitivity        | If $P \subseteq Q$ and $Q \subseteq R$ then $P \subseteq R$                                                  |

### 2.5.1 Crisp Relations

In case of crisp relation  $R$  between two crisp sets  $A$  and  $B$ , two elements,  $x$  from  $A$  and  $y$  from  $B$ , are either related, or not. There is no scope of being partially related to each other. Therefore a crisp relation is defined simply as a subset of the Cartesian product of sets concerned.

**Definition 2.19** (*Cartesian product*) Let  $A$  and  $B$  be two sets. Then the **Cartesian product** of  $A$  and  $B$ , denoted by  $A \times B$ , is the set of all ordered pairs  $(a, b)$  such that  $a \in A$ , and  $b \in B$ .

$$A \times B = \{ (a, b) \mid a \in A, \text{ and } b \in B \} \quad \clubsuit$$

Since  $(a, b) \neq (b, a)$  we have in general  $A \times B \neq B \times A$ . Hence the operation of Cartesian product is not commutative.

**Example 2.13** (*Cartesian product*) Let  $A = \{ p, q, r \}$ , and  $B = \{ 1, 2, 3 \}$ . Then  $A \times B = \{ (p, 1), (p, 2), (p, 3), (q, 1), (q, 2), (q, 3), (r, 1), (r, 2), (r, 3) \}$ , and  $B \times A = \{ (1, p), (1, q), (1, r), (2, p), (2, q), (2, r), (3, p), (3, q), (3, r) \}$ .  $\clubsuit$

**Example 2.14** ( *Cartesian product* ) Let  $A = \{ \text{Math, Phy, Chem} \}$ , and  $B = \{ O, E, A, B, C \}$  is the set of possible grades, e.g., *outstanding* (  $O$  ), *excellent* (  $E$  ), *very good* (  $A$  ), *good* (  $B$  ), and *fair* (  $C$  ). Then  $A \times B = \{ ( \text{Math}, O ), ( \text{Math}, E ), \dots, ( \text{Math}, C ), ( \text{Phy}, O ), \dots, ( \text{Chem}, C ) \}$ . ♣

Obviously, there are as many elements in  $A \times B$  as the product of the number of elements of  $A$  and  $B$ .

$$|A \times B| = |A| \times |B|$$

**Definition 2.20** ( *Crisp relation* ) Given two crisp sets  $A$  and  $B$ , a **crisp relation**  $R$  between  $A$  and  $B$  is a subset of  $A \times B$ .

$$R \subseteq A \times B \quad \clubsuit$$

**Example 2.15** ( *Crisp relation* ) Consider the sets  $A = \{ 1, 2, 3 \}$ ,  $B = \{ 1, 2, 3, 4 \}$  and the relation  $R = \{ ( a, b ) \mid b = a + 1, a \in A, \text{ and } b \in B \}$ . Then  $R = \{ ( 1, 2 ), ( 2, 3 ), ( 3, 4 ) \}$ . Obviously, here  $R \subset A \times B$ . ♣

A crisp relation between sets  $A$  and  $B$  is conveniently expressed with the help of a *relation matrix*  $T$ . The rows and the columns of the relation matrix  $T$  correspond to the members of  $A$  and  $B$  respectively. The entries of  $T$  are defined as

$$T_{ij} = \begin{cases} 1, & \text{if } (a_i, b_j) \in R \\ 0, & \text{otherwise} \end{cases} \quad (2.22)$$

**Example 2.16** ( *Relation matrix for crisp relation* ) Let us, once again, consider the sets  $A = \{ 1, 2, 3 \}$ ,  $B = \{ 1, 2, 3, 4 \}$  and the relation  $R = \{ ( a, b ) \mid b = a + 1, a \in A, \text{ and } b \in B \}$  cited in the previous example. The relation matrix for  $R$  is given below.

$$T_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} & \begin{matrix} \text{column} \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \\ \begin{matrix} \text{row} \end{matrix} & & \end{matrix}$$

♣

**Example 2.17** ( *Relation matrix for crisp relation* ) Let  $P = \{ \text{Tony, Bobby, Mike} \}$  be a set of three students and  $Q = \{ \text{Math, Phy, Chem} \}$  be a set of three subjects in which *Tony*, *Bobby*, and *Mike* have taken a test. TABLE 2.6 shows the grades obtained by these students in these subjects.

**TABLE 2.6.** GRADES OBTAINED BY THREE STUDENTS

|              | <i>Math</i> | <i>Phy</i> | <i>Chem</i> |
|--------------|-------------|------------|-------------|
| <i>Tony</i>  | <i>C</i>    | <i>B</i>   | <i>A</i>    |
| <i>Bobby</i> | <i>A</i>    | <i>A</i>   | <i>B</i>    |
| <i>Mike</i>  | <i>C</i>    | <i>A</i>   | <i>A</i>    |

We define a relation  $R$  between a student and a subject in which they have secured  $A$  grade as  $R = \{ (x, y) \mid x \in P, y \in Q, \text{ and } x \text{ has secured grade } A \text{ in subject } y \}$ . Then  $R = \{ (Tony, Chem), (Bobby, Math), (Bobby, Phy), (Mike, Phy), (Mike, Chem) \}$ . The corresponding relation matrix is shown below.

$$T_R = \begin{matrix} & \begin{matrix} Math & Phy & Chem \end{matrix} \\ \begin{matrix} T \\ B \\ M \end{matrix} & \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

♣

**TABLE 2.7.** OPERATIONS ON CRISP RELATIONS

| # | Operation                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <i>Union</i> ( $R \cup S$ )        | Let $R$ and $S$ are relations defined on $A \times B$ . Then $R \cup S = \{ (a, b) \mid (a, b) \in R, \text{ or } (a, b) \in S, \text{ or both.} \}$ . In terms of the relation matrices, this can be equivalently stated as $(R \cup S)(a, b) = \max \{ R(a, b), S(a, b) \}$ , where $R(a, b)$ , $S(a, b)$ , or $(R \cup S)(a, b)$ are the $(a, b)^{\text{th}}$ element of the relations $R$ , $S$ , or $R \cup S$ respectively. |
| 2 | <i>Intersection</i> ( $R \cap S$ ) | $R \cap S = \{ (a, b) \mid (a, b) \in R, \text{ and } (a, b) \in S \}$ . In other words, $(R \cap S)(a, b) = \min \{ R(a, b), S(a, b) \}$ .                                                                                                                                                                                                                                                                                       |
| 3 | <i>Complementation</i> ( $R'$ )    | $R' = \{ (a, b) \mid (a, b) \notin R \}$ , i.e., $R'(a, b) = 1 - R(a, b)$ .                                                                                                                                                                                                                                                                                                                                                       |
| 4 | <i>Composition</i> ( $R \circ S$ ) | Let $R$ and $S$ are relations defined on $A \times B$ and $B \times C$ respectively. Then $R \circ S = \{ (a, c) \mid (a, c) \in A \times C, \text{ and there exists } b \in B \text{ such that } (a, b) \in R, \text{ and } (b, c) \in S \}$ .                                                                                                                                                                                   |

**Operations on Relations** Certain operations, e.g., *union*, *intersection*, *complementation*, *composition* etc., are occasionally applied on relations. TABLE 2.7 provides the description of these operations with respect to relations  $R$  and  $S$ .

The **Composition** ( $R \circ S$ ) operation is also known as **max-min composition** because it can be equivalently defined, in terms of the corresponding relation matrices, as  $R \circ S = \{ (a, c) \mid (a, c) \in A \times C, \text{ and } (R \circ S)(a, c) = \max [ \min \{ R(a, b), S(b, c) \} ]$ .

**Example 2.18 (Composition)** Let  $A = B = C = \{ 0, 1, 2, 3 \}$  and the relations  $R, S$ , and  $T$  defined as follows :

$$R \subseteq A \times B, R = \{ (a, b) \mid a + b \text{ is an even number} \}$$

$$S \subseteq A \times B, S = \{ (a, b) \mid b = (a + 2) \text{ MOD } 3 \}$$

$$T \subseteq B \times C, T = \{ (b, c) \mid |b - c| = 1 \}$$

These relations can be explicitly written as

$$R = \{ (0, 0), (0, 2), (1, 1), (1, 3), (2, 0), (2, 2), (3, 1), (3, 3) \}$$

$$S = \{ (0, 2), (1, 0), (2, 1), (3, 2) \}, \text{ and}$$

$$T = \{ (0, 1), (1, 0), (1, 2), (2, 1), (2, 3), (3, 2) \}.$$

The relation matrices  $T_R, T_S, T_T$  for  $R, S$ , and  $T$  are given below.

$$T_R = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{array} \quad T_S = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \quad T_T = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Relations  $R \cup S, R \cap S$ , and  $R'$  can be easily obtained directly through the definitions of union, intersection and complementation given above. Their relation matrices  $T_{R \cup S}, T_{R \cap S}$ , and  $T_{R'}$  are given below.

$$T_{R \cup S} = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \end{array} \quad T_{R \cap S} = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \quad T_{R'} = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Let us now compute the composite relation  $R \circ T$ . By definition  $R \circ S$ , the ordered pair  $(0, 0)$  is in  $R \circ T$  if and only if there exists an  $x \in B$  such that  $(0, x) \in R$ , and  $(x, 0) \in T$ . From the relation matrix  $T_R$  of  $R$  we see that there are two ordered pairs,  $(0, 0)$  and  $(0, 2)$  in  $R$  with 0 as the first element. The corresponding second elements are 0, and 2

respectively. Therefore if any of the pairs  $(0, 0)$  and  $(2, 0)$  appears in  $T$ , then  $(0, 0)$  is in  $R \circ T$ . However, a look into the relation matrix of  $T$ , i.e.,  $T_T$ , reveals that neither  $(0, 0)$ , nor  $(2, 0)$  belongs to the relation  $T$ . Hence  $(0, 0) \notin R \circ T$ , and  $T_{R \circ T}(0, 0) = 0$ . On the other hand  $(0, 1) \in R \circ T$  because  $(0, 2) \in R$  and  $(2, 1) \in T$ . Computations of other elements of  $R \circ T$  are shown below.

$$\begin{array}{llll}
(0, 2) \in R & \text{and} & (2, 1) \in T, & \therefore (0, 1) \in R \circ T \\
(0, 2) \in R & \text{and} & (2, 3) \in T, & \therefore (0, 3) \in R \circ T \\
(1, 1) \in R & \text{and} & (1, 0) \in T, & \therefore (1, 0) \in R \circ T \\
(1, 1) \in R & \text{and} & (1, 2) \in T, & \therefore (1, 2) \in R \circ T \\
(2, 0) \in R & \text{and} & (0, 1) \in T, & \therefore (2, 1) \in R \circ T \\
(2, 2) \in R & \text{and} & (2, 3) \in T, & \therefore (2, 3) \in R \circ T \\
(3, 1) \in R & \text{and} & (1, 0) \in T, & \therefore (3, 0) \in R \circ T \\
(3, 1) \in R & \text{and} & (1, 2) \in T, & \therefore (3, 2) \in R \circ T
\end{array}$$

Hence the relation matrix  $T_{R \circ T}$  for  $R \circ T$  looks like

$$T_{R \circ T} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

♣

It should be noted that for crisp relations  $R$  and  $T$  their composition  $R \circ T$  as defined above and their *max-min* composition, to be defined in the next subsection in the context of fuzzy relations, are equivalent. In fact crisp composition is a special case of max-min composition and the later is a generalized concept which is applicable to the realm of fuzzy relations too.

### 2.5.2 Fuzzy Relations

As stated earlier the concept of crisp relations can be generalized to that of *fuzzy relations*. All we have to do is to allow the pairs of elements to be partially related, i.e., the entries of the relation matrix would be anything between 0 and 1. Let us consider some simple instances before the formal definition of fuzzy relations is presented.

**Example 2.19 ( Fuzzy relation )** Let  $P = \{ \text{money, fame, power} \}$  and  $Q = \{ \text{politics, showbiz, academics} \}$  be two crisp sets. Then  $R$  might be an imaginary relation between  $P$  and  $Q$  expressed by the relation matrix

$$T_R = \begin{matrix} & \begin{matrix} ps & sz & as \end{matrix} \\ \begin{matrix} m \\ f \\ p \end{matrix} & \begin{bmatrix} 0.7 & 0.9 & 0.6 \\ 0.8 & 0.9 & 0.5 \\ 1.0 & 0.7 & 0.3 \end{bmatrix} \end{matrix} \quad \begin{matrix} m : \text{money} & ps : \text{politics} \\ f : \text{fame} & sz : \text{showbiz} \\ p : \text{power} & as : \text{academics} \end{matrix}$$

♣

**Example 2.20 ( Fuzzy relation )** Let us consider the set of all real numbers in the interval  $P = [ 0, 1 ] = \{ x \mid 0 \leq x \leq 1 \}$ . A relation  $R$ , say *a close to b*, on  $P^2$  ( i.e.  $P \times P$  ) may be defined in the following way

$$R( a, b ) = 1 - | a - b |, \forall a, b \text{ in } [ 0, 1 ]$$

♣

It should be noted that fuzzy relations defined on discrete sets can be expressed with the help of relation matrices. Relations defined on continuous domains as exemplified in Example 2.20 can not be expressed with any relation matrix. Moreover, in the foregoing discussions we have considered fuzzy relations on crisp sets. It is also possible to define a fuzzy relation on fuzzy sets. This is dependent on the concept of *fuzzy Cartesian product*.

**Definition 2.21 ( Fuzzy Cartesian product )** Let  $A$  and  $B$  be two fuzzy sets on reference sets  $X$  and  $Y$  respectively. The **fuzzy Cartesian product** of  $A$  and  $B$ , denoted by  $A \times B$ , is defined as  $A \times B \subseteq X \times Y$ , and  $\mu_{A \times B}( a, b ) = \min \{ \mu_A( a ), \mu_B( b ) \} \forall a \in X, \forall b \in Y$ .

♣

**Example 2.21 ( Fuzzy Cartesian product )** Let us consider the reference sets  $X = \{ m, n \}$  and  $Y = \{ p, q, r \}$  and the fuzzy sets  $A$  and  $B$  defined on them.

$$A = \frac{0.3}{m} + \frac{0.7}{n}, \quad B = \frac{0.5}{p} + \frac{0.1}{q} + \frac{0.8}{r}$$

Now  $\mu_{A \times B}( m, p ) = \min \{ \mu_A( m ), \mu_B( p ) \} = \min \{ 0.3, 0.5 \} = 0.3$ . But  $\mu_{A \times B}( m, q ) = \min \{ \mu_A( m ), \mu_B( q ) \} = \min \{ 0.3, 0.1 \} = 0.1$ . The membership values of the other elements of the fuzzy Cartesian product  $A \times B$  can also be found in similar fashion. The Cartesian product itself can be looked upon as a fuzzy relation between the fuzzy sets  $A$  and  $B$ . It can be expressed with the help of the relation matrix

$$A \times B = \begin{matrix} & \begin{matrix} p & q & r \end{matrix} \\ \begin{matrix} m \\ n \end{matrix} & \begin{bmatrix} 0.3 & 0.1 & 0.3 \\ 0.5 & 0.1 & 0.7 \end{bmatrix} \end{matrix}$$

♣



A **fuzzy relation**  $R$  between two fuzzy sets  $A$  and  $B$  is a subset of the fuzzy Cartesian product  $A \times B$ . Hence  $R \subseteq A \times B$  where  $A \times B$  is the fuzzy Cartesian product of the fuzzy sets  $A$  and  $B$ .

**Example 2.22 ( Fuzzy relation )** We consider the fuzzy sets  $A$  and  $B$  cited in Example 2.21 and their Cartesian product. The relation matrix given below presents a subset of  $A \times B$  because for all  $x \in X$  and  $y \in Y$ ,  $\mu_R(x, y) \leq \mu_{A \times B}(x, y)$ . Therefore  $R$  is a fuzzy relation between the fuzzy sets  $A$  and  $B$ .

$$R = \begin{matrix} & \begin{matrix} p & q & r \end{matrix} \\ \begin{matrix} m \\ n \end{matrix} & \begin{bmatrix} 0.2 & 0.1 & 0.1 \\ 0.3 & 0.0 & 0.5 \end{bmatrix} \end{matrix}$$

Now let us consider the matrix  $R'$  given below.

$$R' = \begin{matrix} & \begin{matrix} p & q & r \end{matrix} \\ \begin{matrix} m \\ n \end{matrix} & \begin{bmatrix} 0.2 & 0.3 & 0.1 \\ 0.3 & 0.0 & 0.5 \end{bmatrix} \end{matrix}$$

Here  $R'(m, q) = 0.3 > A \times B(m, q) = 0.1$ . Hence  $R'$  is not a fuzzy relation between the fuzzy sets  $A$  and  $B$ . ♣

### 2.5.3 Operations on Fuzzy Relations

The familiar set theoretic operations, e.g., union, intersection, complementation etc. are applicable to fuzzy relations also.

|                 |                                                  |
|-----------------|--------------------------------------------------|
| Union           | $(R \cup S)(x, y) = \max \{ R(x, y), S(x, y) \}$ |
| Intersection    | $(R \cap S)(x, y) = \min \{ R(x, y), S(x, y) \}$ |
| Complementation | $R'(x, y) = 1 - R(x, y)$                         |

Moreover the common relations of inclusion, dominance, and equality also hold good for fuzzy relations.

|           |                                                        |
|-----------|--------------------------------------------------------|
| Inclusion | $R \subseteq S$ if $\forall x, y R(x, y) \leq S(x, y)$ |
| Dominance | $R \supseteq S$ if $\forall x, y R(x, y) \geq S(x, y)$ |
| Equality  | $R = S$ if $\forall x, y R(x, y) = S(x, y)$            |

Apart from the operations and properties stated above the crisp relational operation of composition of two relations is generalized to the so-called *max-min composition* of fuzzy relations.

**Definition 2.21** (*max-min composition*) Let  $A$ ,  $B$ , and  $C$  be three crisp sets.  $R$  and  $S$  are fuzzy relations over  $A \times B$  and  $B \times C$  respectively. The **max-min composition** of  $R$  and  $S$ , denoted by  $R \circ S$ , is a relation over  $A \times C$  such that

$$(R \circ S)(x, y) = \max_{y \in B} \{ \min(R(x, y), S(y, z)) \} \quad \clubsuit$$

**Example 2.23** (*max-min composition*) Suppose  $A = \{ a, b, c \}$ ,  $B = \{ x, y \}$ , and  $C = \{ p, q, r \}$  be three crisp sets. There are two fuzzy relations  $R$  and  $S$  defined over  $A \times B$  and  $B \times C$  respectively. The relation matrices of  $R$  and  $S$  are

$$R = \begin{matrix} & \begin{matrix} x & y \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0.3 & 0.7 \\ 0.9 & 0.4 \\ 0.2 & 0.5 \end{bmatrix} \end{matrix}, \quad S = \begin{matrix} & \begin{matrix} p & q & r \end{matrix} \\ \begin{matrix} x \\ y \end{matrix} & \begin{bmatrix} 0.4 & 0.1 & 0.8 \\ 0.3 & 0.7 & 0.6 \end{bmatrix} \end{matrix}$$

The max-min composition  $R \circ S$  is defined on the Cartesian product  $A \times C$ . Let us consider the computation of the first element  $(R \circ S)(a, p)$ .

$$\begin{aligned} (R \circ S)(a, p) &= \max_{i \in B} \{ \min(R(a, i), S(i, p)) \} \\ &= \max \{ \min(R(a, x), S(x, p)), \min(R(a, y), S(y, p)) \} \\ &= \max \{ \min(0.3, 0.4), \min(0.7, 0.3) \} \\ &= \max \{ 0.3, 0.3 \} \\ &= 0.3 \end{aligned}$$

Similarly, the next element is computed as follows.

$$\begin{aligned} (R \circ S)(a, q) &= \max_{i \in B} \{ \min(R(a, i), S(i, q)) \} \\ &= \max \{ \min(R(a, x), S(x, q)), \min(R(a, y), S(y, q)) \} \\ &= \max \{ \min(0.3, 0.1), \min(0.7, 0.7) \} \\ &= \max \{ 0.1, 0.7 \} \\ &= 0.7 \end{aligned}$$

Computation of the rest of the elements is left as an exercise. Finally  $R \circ S$  looks like

$$R \circ S = \begin{matrix} & \begin{matrix} p & q & r \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0.3 & 0.7 & 0.6 \\ 0.4 & 0.7 & 0.8 \\ 0.3 & 0.5 & 0.5 \end{bmatrix} \end{matrix}$$

♣

**Example 2.24 ( max-min composition )** Let us consider two kinds of troubles a PC may suffer from, viz., the *system hangs while running*, and *the system does not boot*. We symbolize the former by  $h$  and the later by  $b$  and define the set  $A = \{ h, b \}$  of PC troubles. Two possible causes of these troubles are *computer virus* ( $v$ ) and *disc crash* ( $c$ ) and they form the set  $B = \{ c, v \}$  PC trouble makers. And finally, let the sources of the causes mentioned above are *internet* ( $i$ ) and *obsolescence* ( $o$ ) and  $C = \{ i, o \}$  is the set of PC trouble causes. The relation between PC troubles and their causes is expressed by  $R$ , a fuzzy relation over  $A \times B$ . Similarly,  $S$  is the fuzzy relation over  $B \times C$ , i.e., the relation between the causes of troubles and the sources of those causes. The relations  $R$  and  $S$  in terms of their relation matrices are shown below.

$$R = \begin{matrix} & \begin{matrix} v & c \end{matrix} \\ \begin{matrix} h \\ b \end{matrix} & \begin{bmatrix} 0.7 & 0.2 \\ 0.5 & 0.8 \end{bmatrix} \end{matrix}, \quad S = \begin{matrix} & \begin{matrix} i & o \end{matrix} \\ \begin{matrix} v \\ c \end{matrix} & \begin{bmatrix} 0.9 & 0.7 \\ 0.1 & 0.2 \end{bmatrix} \end{matrix}$$

The relation between PC troubles and their ultimate sources, i.e., between  $A$  and  $C$ , can be computed on the basis of  $R$  and  $S$  above as the max-min composition  $R \circ S$ . The first element of  $R \circ S$ , expressed as  $(R \circ S)(h, i)$  is computed as follows.

$$\begin{aligned} (R \circ S)(h, i) &= \max \{ \min (R(h, v), S(v, i)), \min (R(h, c), S(c, i)) \} \\ &= \max \{ \min (0.7, 0.9), \min (0.2, 0.1) \} \\ &= \max \{ 0.7, 0.1 \} \\ &= 0.7 \end{aligned}$$

The rest of the elements of  $R \circ S$  can be found in a similar fashion.

$$\begin{aligned} (R \circ S)(h, o) &= 0.7 \\ (R \circ S)(b, i) &= 0.5 \\ (R \circ S)(b, o) &= 0.5 \end{aligned}$$

And finally we get,

$$R \circ S = \begin{matrix} & \begin{matrix} i & o \end{matrix} \\ \begin{matrix} h \\ b \end{matrix} & \begin{bmatrix} 0.7 & 0.7 \\ 0.5 & 0.5 \end{bmatrix} \end{matrix}$$

♣

## 2.6 Fuzzy Extension Principle

This section presents a discussion on the fuzzy extension principle which provides a way to map certain mathematical concepts of crisp set theory to their fuzzy counterparts.

### 2.6.1 Preliminaries

Before we present the fuzzy extension principle, it is necessary to understand a few concepts that form the basis of the said principle. This subsection provides the groundwork for this purpose.

**a) Level Set** Corresponding to every fuzzy set there is a crisp set consisting of the membership values of its singletons. This is known as the *level set* of the fuzzy set. Its members are real values between 0 and 1, including 1 but excluding 0.

**Definition 2.22 (Level set)** Let  $F$  be a fuzzy set on the universe  $U$ . The **level set** of  $F$ , denoted as  $L(F)$ , is a crisp set of real values  $x \in (0, 1]$  such that for each  $x \in L(F)$  there is a singleton  $(y, x) \in F$ .

$$L(F) = \{ x \mid 0 < x \leq 1 \text{ and } \exists y \in U \text{ such that } \mu_F(y) = x \} \quad \clubsuit$$

**Example 2.25 (Level set)** Let  $F$  be a fuzzy set on the universe  $U = \{ a, b, c, d, e \}$ .

$$F = \frac{0.3}{a} + \frac{0.8}{b} + \frac{0.0}{c} + \frac{0.5}{d} + \frac{0.7}{e}$$

The corresponding level set is  $L(F) = \{ 0.3, 0.5, 0.7, 0.8 \}$ . It may be noted that the membership value 0.0 of the element  $c$  is not included in  $L(F)$ . ♣

**Example 2.26 (Level set)** Let  $F$  be a fuzzy set on the universe  $U = \{ a, b, c, d, e \}$ .

$$F = \frac{0.3}{a} + \frac{0.8}{b} + \frac{0.0}{c} + \frac{0.3}{d} + \frac{0.7}{e}$$

The corresponding level set is  $L(F) = \{ 0.3, 0.7, 0.8 \}$ . The membership value 0.3 is repeated twice in  $F$  so that there are only three distinct non-zero membership values included in  $L(F)$ . ♣

**Example 2.27 (Level set)** Let us consider the following membership function for a fuzzy set  $F$ .

$$\mu_F(x) = \begin{cases} 1 - e^{-x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

If the universe of discourse is the entire real line extending from  $-\infty$  to  $+\infty$  then the membership function will attain values within the range  $[0, 1]$ . Hence  $L(F) = [0, 1]$ . ♣

**b)  $\alpha$ -cut** Occasionally we are interested in the elements whose degrees of membership to a given fuzzy set lie above a given level. This is provided with the help of the  $\alpha$ -cut.

**Definition 2.23 (  $\alpha$ -cut )** Let  $F$  be a fuzzy set on the universe  $U$  and  $\alpha$  be a number such that  $0 < \alpha \leq 1$ . The  $\alpha$ -cut of  $F$ , denoted as  $F_\alpha$ , is a crisp set containing those elements of  $U$  whose membership value with respect to the fuzzy set  $F$  is greater than, or equal to,  $\alpha$ .

$$F_\alpha = \{ x \in U \mid \mu_F(x) \geq \alpha \} \quad \clubsuit$$

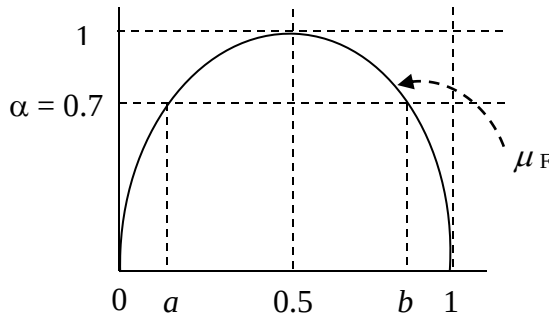
**Example 2.28 (  $\alpha$ -cut )** Let  $U = \{ a, b, c, d \}$  be a universe and  $F$  be a fuzzy set on  $U$ .

$$F = \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.7}{c} + \frac{1.0}{d}$$

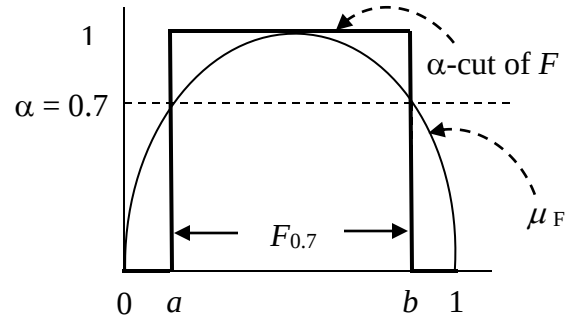
The  $\alpha$ -cuts for various  $\alpha$  are :  $F_{1.0} = \{ d \}$ ,  $F_{0.7} = \{ c, d \}$ ,  $F_{0.6} = \{ a, c, d \}$ ,  $F_{0.3} = \{ a, b, c, d \}$ . Moreover, it is obvious from the example that

$$\begin{aligned} \text{For all } \alpha, 0 < \alpha \leq 0.3, F_\alpha &= F_{0.3} \\ \text{For all } \alpha, 0.3 < \alpha \leq 0.6, F_\alpha &= F_{0.6} \\ \text{For all } \alpha, 0.6 < \alpha \leq 0.7, F_\alpha &= F_{0.7} \\ \text{For all } \alpha, 0.7 < \alpha \leq 1.0, F_\alpha &= F_{1.0} \end{aligned} \quad \clubsuit$$

**Example 2.29 (  $\alpha$ -cut )** Let  $F$  be a fuzzy set defined on  $U = [0, 1]$ . The membership function  $\mu_F$  is graphically shown in Fig. 2.28. Fig. 2.29 shows the profile of the  $\alpha$ -cut of  $F$  for  $\alpha = 0.7$ . ♣



**Fig. 2.28**



**Fig. 2.29**

It may be noted that as  $\alpha$  increases from 0 to 1, the size of the corresponding  $\alpha$ -cuts decreases, i.e., given two real values  $\alpha$  and  $\beta$ ,  $0 < \alpha \leq \beta \leq 1$ , we must have  $F_\alpha \supseteq F_\beta$ .

**c) The  $\alpha$ -cut decomposition theorem** Before introducing the  $\alpha$ -cut decomposition theorem we need to get acquainted with a new notation. Given a crisp set  $A$ , and a real number  $x$ ,  $0 \leq x \leq 1$ ,  $x.A$  is a fuzzy set consisting of all members of  $A$  with membership value  $x$ . For example, if  $A = \{ p, q, r \}$  and  $x$  is 0.4, then  $x.A = \{ (p, 0.4), (q, 0.4), (r, 0.4) \}$ .

The  **$\alpha$ -cut decomposition theorem** states that any fuzzy set  $F$  can be decomposed into a number of the  $\alpha$ -cuts, say  $F_\alpha$ s, such that

$$F = \bigcup_{\alpha} \alpha.F_{\alpha}$$

**Example 2.30 (  $\alpha$ -cut decomposition theorem )** Let  $F = \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.7}{c} + \frac{1.0}{d}$  be a fuzzy set on the universe  $U = \{ a, b, c, d \}$ . We see

$$\begin{aligned} F_{0.3} &= \{ a, b, c, d \}, & \text{and} & & 0.3.F_{0.3} &= \frac{0.3}{a} + \frac{0.3}{b} + \frac{0.3}{c} + \frac{0.3}{d} \\ F_{0.6} &= \{ a, c, d \}, & \text{and} & & 0.6.F_{0.6} &= \frac{0.6}{a} + \frac{0.6}{c} + \frac{0.6}{d} \\ F_{0.7} &= \{ c, d \}, & \text{and} & & 0.7.F_{0.7} &= \frac{0.7}{c} + \frac{0.7}{d} \\ F_{1.0} &= \{ d \}, & \text{and} & & 1.0.F_{1.0} &= \frac{1.0}{d} \end{aligned}$$

$$\text{Therefore } (0.3.F_{0.3}) \cup (0.6.F_{0.6}) \cup (0.7.F_{0.7}) \cup (1.0.F_{1.0}) = \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.7}{c} + \frac{1.0}{d} = F \clubsuit$$

**d) Restricted scalar multiplication ( RSM )** Restricted scalar multiplication ( RSM ) is a way of relating two fuzzy sets. Given a fuzzy set  $F$  on the universe  $U$ , and a real number  $\alpha$ ,  $0 \leq \alpha \leq 1$ , **restricted scalar multiplication ( RSM )** of  $F$  by  $\alpha$  is the process of creating another set, denoted by  $\alpha F$  on the same universe whose memberships are obtained as

$$\mu_{\alpha F}(x) = \alpha \cdot \mu_F(x), \forall x \in U$$

RSM is a kind of fuzzification procedure because it produces a fuzzy set out of a given set, crisp, or fuzzy.

**Example 2.31 ( Restricted scalar multiplication )** Consider the fuzzy set  $F = \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.7}{c} + \frac{1.0}{d}$  on the universe  $U = \{ a, b, c, d \}$  cited in Example 2.30. If  $\alpha = 0.1$ . Then RSM of  $F$  by  $\alpha$  produces the set  $\alpha F = \frac{0.06}{a} + \frac{0.03}{b} + \frac{0.07}{c} + \frac{0.1}{d}$ .  $\clubsuit$

Like other operations on fuzzy sets *RSM* too satisfies certain properties. Given the fuzzy sets  $F$  and  $G$  on the universe  $U$ , and two real numbers  $\alpha, \beta, 0 < \alpha, \beta \leq 1$ , the following properties hold good.

- i)  $\alpha (F \cup G) = \alpha F \cup \alpha G$
- ii)  $\alpha (F \cap G) = \alpha F \cap \alpha G$
- iii)  $(\alpha \beta) F = \alpha (\beta F)$
- iv)  $1 \cdot F = F$
- v)  $\alpha F \subseteq F$

**Definition 2.24** (*Fuzzy cardinality*) Let  $F$  be a fuzzy set on the universe  $U$ .  $L(F)$  is the level set of  $F$  and for all  $\alpha \in L(F)$ ,  $F_\alpha$  is the  $\alpha$ -cut of  $F$ . The **fuzzy cardinality** of  $F$ , denoted by  $f_c(F)$ , is defined as the fuzzy set

$$f_c(F) = \sum_{\alpha \in L(F)} \frac{\alpha}{|F_\alpha|} \quad \clubsuit$$

$|F_\alpha|$  is the cardinality of  $F_\alpha$  which is a natural number. Therefore, fuzzy cardinality of any fuzzy set is another fuzzy set defined on natural numbers. The following example illustrates the concept of fuzzy cardinality.

**Example 2.32** (*Fuzzy cardinality*) Let us consider the fuzzy set  $H$  of four happy persons.

$$H = \frac{0.3}{mita} + \frac{0.7}{neeta} + \frac{0.4}{jeet} + \frac{0.8}{joy}$$

We have the level set  $L(H) = \{0.3, 0.4, 0.7, 0.8\}$  and the following cardinalities

$$\begin{aligned} |H_{0.3}| &= |\{mita, neeta, jeet, joy\}| = 4, \\ |H_{0.4}| &= |\{neeta, jeet, joy\}| = 3, \\ |H_{0.7}| &= |\{neeta, joy\}| = 2, \\ |H_{0.8}| &= |\{joy\}| = 1. \end{aligned}$$

Hence the fuzzy cardinality of  $H$  is given by

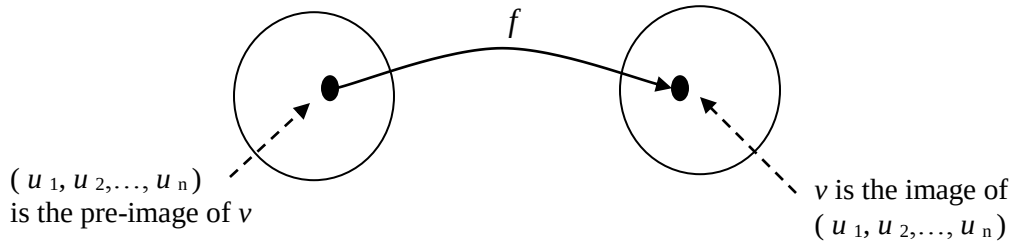
$$f_c(H) = \sum_{\alpha \in L(H)} \frac{\alpha}{|H_\alpha|} = \frac{0.3}{4} + \frac{0.4}{3} + \frac{0.7}{2} + \frac{0.8}{1} \quad \clubsuit$$

## 2.6.2 The Extension Principle

The extension principle provides a way to map certain mathematical concepts pertaining to crisp sets to their fuzzy counterparts.

Consider a function  $f: U_1 \times U_2 \times \dots \times U_n \rightarrow V$  where the  $U_i$ s and  $V$  are crisp sets and  $U_1 \times U_2 \times \dots \times U_n$  is the Cartesian product of  $U_1, U_2, \dots, U_n$ . If there is a point  $(u_1, u_2, \dots, u_n)$  in the  $n$ -dimensional space  $U_1 \times U_2 \times \dots \times U_n$  and there is a  $v \in V$  such that  $f(u_1, u_2, \dots, u_n) = v$  then  $v$  is said to be the *image* of the point  $(u_1, u_2, \dots, u_n)$ . Equivalently, the point  $(u_1, u_2, \dots, u_n)$  is referred to as the *pre-image* of  $v$ , ( see Fig. 2.30 ) and is indicated by the expression

$$(u_1, u_2, \dots, u_n) = f^{-1}(v)$$



**Fig. 2.30.** An image and its pre-image

Now, let us consider fuzzy sets  $A_1, A_2, \dots, A_n$  on the universes  $U_1, U_2, \dots, U_n$  respectively. Given the function  $f$  as mentioned above it is possible to construct a fuzzy set  $B$  on  $V$  in the following manner:

for each  $v \in V$  we define

$$\mu_B(v) = 0, \text{ if } f^{-1}(v) = \phi, \text{ i.e., there is no pre-image of } v \text{ in } U_1 \times U_2 \times \dots \times U_n$$

$$= \max_{\substack{\forall (u_1, u_2, \dots, u_n), \\ (u_1, u_2, \dots, u_n) = f^{-1}(v)}} \{ \min(\mu_{A_1}(u_1), \mu_{A_2}(u_2), \dots, \mu_{A_n}(u_n)) \} \text{ otherwise.}$$

If we define a fuzzy set  $A$  on  $A_1 \times A_2 \times \dots \times A_n$  where  $\mu_A(u_1, u_2, \dots, u_n) = \min \{ \mu_{A_1}(u_1), \mu_{A_2}(u_2), \dots, \mu_{A_n}(u_n) \}$  then the procedure described above to obtain the fuzzy set  $B$  can be considered to be an extension of the crisp function  $f: U_1 \times U_2 \times \dots \times U_n \rightarrow V$  to the corresponding fuzzy domain  $f: A_1 \times A_2 \times \dots \times A_n \rightarrow B$ . Here the fuzzy set  $B$  is termed as the image of  $A$  under  $f$ , and is denoted as  $B = f(A)$ . The subsequent two examples illustrate the extension principle described above.

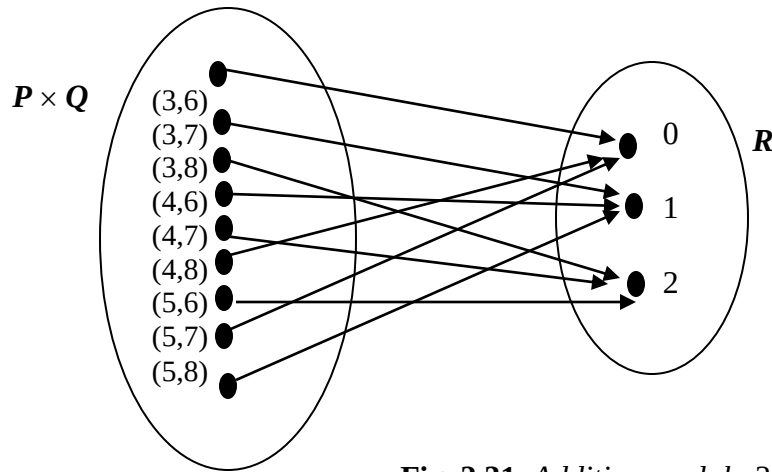
**Example 2.33 ( Fuzzy extension principle )** Let us consider the crisp domains  $P = \{ 3, 4, 5 \}$ ,  $Q = \{ 6, 7, 8 \}$ , and  $R = \{ 0, 1, 2 \}$ . TABLE 2.8 shows the function  $f: P \times Q \rightarrow R$  where  $f$  is defined as *addition modulo 3*. Fig. 2.31 depicts the function graphically.

We see that  $f(3, 6) = f(5, 7) = f(4, 8) = 0$  so that 0 has three pre-images,  $(3, 6)$ ,  $(5, 7)$ , and  $(4, 8)$ . Therefore  $f^{-1}(0) = \{ (3, 6), (5, 7), (4, 8) \}$ . Similarly,  $f^{-1}(1) = \{ (4, 6), (3, 7), (5, 8) \}$  and  $f^{-1}(2) = \{ (5, 6), (4, 7), (3, 8) \}$ .



**TABLE 2.8.** THE FUNCTION  $f \equiv \text{Addition modulo } 3$

|              |          |                                         |          |          |
|--------------|----------|-----------------------------------------|----------|----------|
|              |          | $\longleftrightarrow Q \longrightarrow$ |          |          |
|              |          | <b>6</b>                                | <b>7</b> | <b>8</b> |
| $\uparrow$   | <b>3</b> | 0                                       | 1        | 2        |
| $P$          | <b>4</b> | 1                                       | 2        | 0        |
| $\downarrow$ | <b>5</b> | 2                                       | 0        | 1        |



**Fig. 2.31.** Addition modulo 3 over  $P \times Q$

Now consider the fuzzy sets  $A$  and  $B$  on  $P$  and  $Q$  respectively as described below.

$$A = \frac{0.1}{3} + \frac{0.8}{4} + \frac{0.5}{5}, \quad B = \frac{0.6}{6} + \frac{0.2}{7} + \frac{0.7}{8}$$

Applying the procedure of extension principle, it is possible to *extend* the function  $f \equiv \text{Addition modulo } 3$  as *Fuzzy addition modulo 3* from  $A \times B$  to  $C$ , where  $C$  is a fuzzy set defined on  $R$ . The membership values of various  $x \in R$  in the fuzzy set  $C$  are obtained as described below

a) For  $0 \in R$

$$\begin{aligned}
 f^{-1}(0) &= \{ (3, 6), (5, 7), (4, 8) \} \\
 \therefore \mu_C(0) &= \max \{ \min(\mu_A(3), \mu_B(6)), \min(\mu_A(5), \mu_B(7)), \\
 &\quad \min(\mu_A(4), \mu_B(8)) \} \\
 &= \max \{ \min(0.1, 0.6), \min(0.5, 0.2), \min(0.8, 0.7) \} \\
 &= \max \{ 0.1, 0.2, 0.7 \} \\
 &= 0.7.
 \end{aligned}$$

b) For  $1 \in R$

$$\begin{aligned}
 f^{-1}(1) &= \{ (4, 6), (3, 7), (5, 8) \} \\
 \therefore \mu_C(1) &= \max \{ \min(\mu_A(4), \mu_B(6)), \min(\mu_A(3), \mu_B(7)), \\
 &\quad \min(\mu_A(5), \mu_B(8)) \} \\
 &= \max \{ \min(0.8, 0.6), \min(0.1, 0.2), \min(0.5, 0.7) \} \\
 &= \max \{ 0.6, 0.1, 0.5 \} \\
 &= 0.6.
 \end{aligned}$$

c) For  $2 \in R$

$$\begin{aligned}
 f^{-1}(2) &= \{ (5, 6), (4, 7), (3, 8) \} \\
 \therefore \mu_C(2) &= \max \{ \min(\mu_A(5), \mu_B(6)), \min(\mu_A(4), \mu_B(7)), \\
 &\quad \min(\mu_A(3), \mu_B(8)) \} \\
 &= \max \{ \min(0.5, 0.6), \min(0.8, 0.2), \min(0.1, 0.7) \} \\
 &= \max \{ 0.5, 0.2, 0.1 \} \\
 &= 0.5.
 \end{aligned}$$

Hence the image of  $A \times B$  under  $f$  is given by  $C = \{ (0, 0.7), (1, 0.6), (2, 0.5) \}$ , or

$$C = \frac{0.7}{0} + \frac{0.6}{1} + \frac{0.5}{2}$$

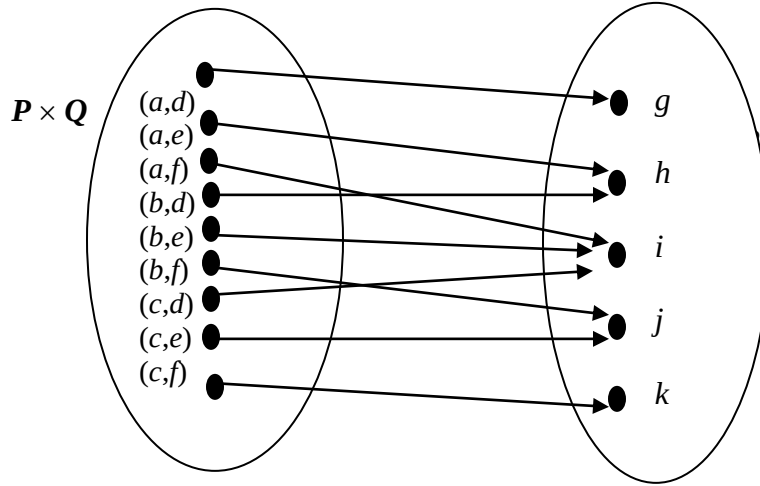
♣

**Example 2.34 (Fuzzy extension principle)** Let  $P = \{ a, b, c \}$ ,  $Q = \{ d, e, f \}$ , and  $R = \{ g, h, i, j, k \}$  be three crisp sets, and  $f: P \times Q \rightarrow R$  be a function defined as  $f(x, y) = x + y + 2$ , where '+' is to be interpreted as the addition of the sequence number of  $x$  and  $y$  in the English alphabet. For example, the letter  $a$  has the sequence number 1 as it is the first letter of the English alphabet, and that of the letter  $d$  is 4. Therefore the expression  $a + d$  produces the letter with sequence number  $1 + 4 = 5$ , the letter  $e$ . Similarly,  $a + d + 2 = g$ . TABLE 2.9 shows the function table for the function  $f: P \times Q \rightarrow R$ . Fig. 2.32 shows the same function graphically.

**TABLE 2.9.** FUNCTION TABLE OF  $f(x, y) = x + y + 2$

|                                   |     | $\longleftrightarrow Q \longrightarrow$ |     |     |
|-----------------------------------|-----|-----------------------------------------|-----|-----|
|                                   |     | $d$                                     | $e$ | $f$ |
| $\uparrow$<br>$P$<br>$\downarrow$ | $a$ | $g$                                     | $h$ | $i$ |
|                                   | $b$ | $h$                                     | $i$ | $j$ |
|                                   | $c$ | $i$                                     | $j$ | $k$ |

Now, let us consider the fuzzy sets  $A$  and  $B$  on  $P$  and  $Q$  as described below.



**Fig. 2.32.**  $f(x, y) = x + y + 2$  over  $P \times Q$

$$A = \frac{0.4}{a} + \frac{0.3}{b} + \frac{0.8}{c}, \quad B = \frac{0.7}{d} + \frac{0.5}{e} + \frac{0.8}{f}$$

Computation of the fuzzy set  $C$  on  $R$  as the image of  $A \times B$  with the extension principle is detailed below.

a) For  $g \in R$

$$\begin{aligned} f^{-1}(g) &= \{ (a, d) \} \\ \therefore \mu_C(g) &= \max \{ \min(\mu_A(a), \mu_B(d)) \} \\ &= \max \{ \min(0.4, 0.7) \} \\ &= \max \{ 0.4 \} \\ &= 0.4. \end{aligned}$$

b) For  $h \in R$

$$\begin{aligned} f^{-1}(h) &= \{ (a, e), (b, d) \} \\ \therefore \mu_C(h) &= \max \{ \min(\mu_A(a), \mu_B(e)), \min(\mu_A(b), \mu_B(d)) \} \\ &= \max \{ \min(0.4, 0.5), \min(0.3, 0.7) \} \\ &= \max \{ 0.4, 0.3 \} \\ &= 0.4. \end{aligned}$$

c) For  $i \in R$

$$\begin{aligned} f^{-1}(i) &= \{ (a, f), (b, e), (c, d) \} \\ \therefore \mu_C(i) &= \max \{ \min(\mu_A(a), \mu_B(f)), \min(\mu_A(b), \mu_B(e)), \\ &\quad \min(\mu_A(c), \mu_B(d)) \} \\ &= \max \{ \min(0.4, 0.8), \min(0.3, 0.5), \min(0.8, 0.7) \} \end{aligned}$$

$$= \max \{ 0.4, 0.3, 0.7 \}$$

$$= 0.7.$$

d) For  $j \in R$

$$f^{-1}(j) = \{ (b, f), (c, e) \}$$

$$\therefore \mu_C(j) = \max \{ \min(\mu_A(b), \mu_B(f)), \min(\mu_A(c), \mu_B(e)) \}$$

$$= \max \{ \min(0.3, 0.8), \min(0.8, 0.5) \}$$

$$= \max \{ 0.3, 0.5 \}$$

$$= 0.5.$$

e) For  $k \in R$

$$f^{-1}(k) = \{ (c, f) \}$$

$$\therefore \mu_C(k) = \max \{ \min(\mu_A(c), \mu_B(f)) \}$$

$$= \max \{ \min(0.8, 0.8) \}$$

$$= \max \{ 0.8 \}$$

$$= 0.8.$$

Hence the image of  $A \times B$  under  $f$  is given by  $C = f(A \times B) = \{ (g, 0.4), (h, 0.4), (i, 0.7), (j, 0.5), (k, 0.8) \}$ , i.e.,

$$C = \frac{0.4}{g} + \frac{0.4}{h} + \frac{0.7}{i} + \frac{0.5}{j} + \frac{0.8}{k}.$$



## Chapter Summary

A summary of the matters discussed in this chapter is provided below.

- Traditional set theory is not adequately equipped to model the vagueness / inexactness that we are used to tackle in our everyday life.
- Fuzzy set theory is an extension, or generalization, of crisp set theory that takes into account the vagueness mentioned above by allowing partial membership to set. Hence, the degree of membership of an element to a fuzzy set is any real value between 0 and 1, both inclusive.
- The membership profile of a fuzzy set is customarily expressed with the help of a membership function  $\mu : U \rightarrow [0, 1]$ . Popular membership functions are the triangular function, trapezoidal function, Gaussian function, S-function etc.
- Fuzzy sets are characterized with the help of certain parameters, e.g., normality, height, support, core, cardinality etc.
- Fuzzy sets are occasionally transformed through operations like normalization, dilation, concentration, contrast intensification, and fuzzyfication. These transformations help to deal with linguistic variables.
- Familiar crisp set operations, e.g., union, intersection, complementation, equality, inclusion, difference, disjunctive sum etc., are extended to fuzzy domain.
- Most of the properties of crisp sets, e.g., associativity, commutativity, distributivity, idempotency, De Morgan's law, transitivity etc., are also satisfied by fuzzy set operations. However, unlike crisp sets, in the fuzzy domain we have  $P \cup P' \neq U$ , and  $P \cap P' \neq \Phi$  because  $\mu_P(x) \in [0, 1]$ , and  $\mu_{P \cup P'}(x) = \max \{ \mu_P(x), 1 - \mu_P(x) \} \neq 1$ , and  $\mu_{P \cap P'}(x) = \min \{ \mu_P(x), 1 - \mu_P(x) \} \neq 0$ .
- The concept of a crisp relation is generalized to fuzzy relation. A fuzzy relation may exist between two crisp sets, or two fuzzy sets. A fuzzy relation between two fuzzy sets is a subset of their fuzzy Cartesian product.
- A fuzzy relation between two crisp sets consists of ordered pairs of the elements of the sets and a membership value for each ordered pair that ranges from 0 to 1.
- Common operations on fuzzy relations are union, intersection, complementation, and max-min composition.
- The fuzzy extension principle provides a way to map certain mathematical concepts pertaining to crisp sets to their fuzzy counterparts.

## Bibliography & Historical Notes

Fuzzy Sets, as we know them today was developed by the noted Iranian scientist Lotfi Akser Zadeh and presented through his celebrated work “Fuzzy Sets” in *Information and Control* (8) 1965, pp. 338-353, as an extension of the classical set theory, hence forth known as crisp sets. However, the concept of fuzziness dates back to 500 B.C when first references are found in teachings of Gautam Buddha, who believed that almost everything in this universe contains a set of opposites. The same was argued by Greek philosopher Heraclitus who believed that things can be simultaneously 'true' and 'not true'. The Greeks toyed with such ideas until Aristotle and his two valued logic gained ground. Around the same time as Zadeh, Dieter Klaua also proposed a similar concept (Klaua, D. (1965) Über einen Ansatz zur mehrwertigen Mengenlehre. Monatsb. Deutsch. Akad. Wiss. Berlin 7, 859–867). Like all revolutionary notions, the concept of fuzzy sets too was viewed initially with lot of skepticism and many rated it merely as an extension of probability theory. It wasn't until the early 1980's that it gained popular ground, and it was around this time that the use of fuzzy controllers in consumer electronic goods started and the word fuzzy became popular among scientists and non-scientists alike. Numerous brilliant works are reported on fuzzy set theory in the past few decades. Only a few of these are cited below.

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## Solved Problems

**Problem 2.1** Which of the following sets are identical to each other?

$$A = \{ x \mid x \text{ is even and } 0 \leq x \leq 10 \}$$

$$B = \{ 0, 2, 4, 6, 8, 10 \}$$

$$C = \{ 10, 0, 8, 2, 6, 4 \}$$

$$D = \{ 6, 8, 6, 0, 0, 4, 10, 10, 2 \}$$

**Solution 2.1** All of the sets  $A$ ,  $B$ ,  $C$  and  $D$  have the same members, irrespective of their descriptions, ordering, or repetition. Hence all of them are identical.  $A = B = C = D$ . ♣

**Problem 2.2** Describe the following sets by citing a property shared by all its members.

$$A = \{ \text{Helium, Neon, Argon, Krypton, Xenon, Radon} \}$$

$$B = \{ 2, 3, 5, 7, 11, 13, 17, 19, 23 \}$$

$$C = \{ 0, 3, 8, 15, 24, 35, 48, 63, \dots \}$$

**Solution 2.2** The required descriptions are given below.

$$A = \{ x \mid x \text{ is an inert gas} \}$$

$$B = \{ x \mid x \text{ is a prime number and } 2 \leq x < 25 \}$$

$$C = \{ x^2 - 1 \mid x \text{ is a natural number} \}$$
 ♣

**Problem 2.3** Let  $A$  be the set of all non-negative integers,  $B$  be that of the non-positive integers inclusive of 0 and  $I$  be the set of all integers, the universal set. Find  $A \cup B$ ,  $A \cap B$ ,  $A'$ ,  $B'$ ,  $A - B$ ,  $B - A$ , and  $A \oplus B$ .

**Solution 2.3** According to the problem statement,  $A = \{ 0, 1, 2, 3, \dots \}$ ,  $B = \{ 0, -1, -2, -3, \dots \}$ , and  $I = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$ . Therefore

$$A \cup B = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \} = I$$

$$A \cap B = \{ 0 \}$$

$$A' = \{ -1, -2, -3, \dots \}$$

$$B' = \{ 1, 2, 3, \dots \}$$

$$A - B = \{ 1, 2, 3, \dots \} = B'$$

$$B - A = \{ -1, -2, -3, \dots \} = A'$$

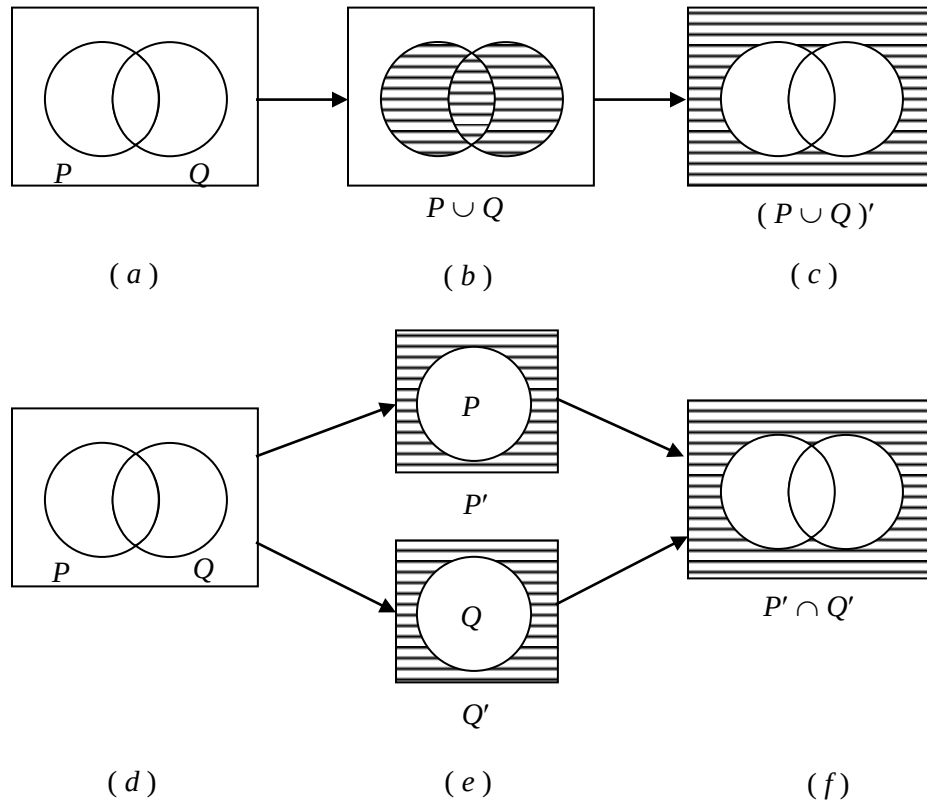
$$A \oplus B = \{ \dots, -3, -2, -1, 1, 2, 3, \dots \}$$
 ♣

**Problem 2.4** For arbitrary sets  $P$  and  $Q$ , under what condition  $P \times Q = Q \times P$ ?

**Solution 2.4** By definition,  $P \times Q = \{ (x, y) \mid x \in P, \text{ and } y \in Q \}$  and  $Q \times P = \{ (x, y) \mid x \in Q, \text{ and } y \in P \}$ . Hence  $P \times Q = Q \times P$  if and only if  $P = Q$ . ♣

**Problem 2.5** Prove De Morgan's theorem with the help of Venn diagrams.

**Solution 2.5** Fig. 2.33 (a) – (f) depict the process of constructing the sets  $(P \cup Q)'$  and  $P' \cap Q'$ . It is seen the Venn diagrams for  $(P \cup Q)'$  and  $P' \cap Q'$  are identical. Hence  $(P \cup Q)' = P' \cap Q'$ , as stated by De Morgan's theorem. The other part of the theorem, i.e.,  $(P \cap Q)' = P' \cup Q'$  can be proved similarly.



**Fig. 2.33.** Proof of De Morgan's theorem

**Problem 2.6** Prove that for any set  $S$ ,  $S \oplus S = \emptyset$ .

**Solution 2.6**  $S \oplus S = (S \cap S') \cup (S' \cap S) = (\emptyset \cup \emptyset) = \emptyset$  ♣



**Problem 2.7.** Prove that  $A - (A - B) = A \cap B$

$$\begin{aligned}
 \text{Solution 2.7} \quad \text{L.H.S} &= A - (A - B) \\
 &= A - (A \cap B') \\
 &= A \cap (A \cap B')' \\
 &= A \cap (A' \cup B) \\
 &= (A \cap A') \cup (A \cap B) \\
 &= \emptyset \cup (A \cap B) \\
 &= A \cap B \\
 &= \text{R.H.S}
 \end{aligned}$$

♣

**Problem 2.8** Prove that if  $A \cup C = B \cup C$ , and  $A \cap C = B \cap C$ , then  $A = B$ . However,  $A \cup C = B \cup C$  does not imply that  $A = B$ . Nor  $A \cap C = B \cap C$  implies that  $A = B$ .

$$\begin{aligned}
 \text{Solution 2.8} \quad A &= A \cup (A \cap C) \\
 &= A \cup (B \cap C) && : \text{Since } A \cap C = B \cap C \text{ ( given )} \\
 &= (A \cup B) \cap (A \cup C) \\
 &= (A \cup B) \cap (B \cup C) && : \text{Since } A \cup C = B \cup C \text{ ( given )} \\
 &= (B \cup A) \cap (B \cup C) \\
 &= B \cup (A \cap C) \\
 &= B \cup (B \cap C) && : \text{Since } A \cap C = B \cap C \text{ ( given )} \\
 &= B
 \end{aligned}$$

♣

**Problem 2.9** Let  $F$  be a fuzzy set of *matured* persons where the maturity is measured in terms of age in years. The fuzzy membership function followed is given below

$$\mu_F(x) = \begin{cases} 0, & \text{if } x \leq 5 \\ \left(\frac{x-5}{20}\right)^2, & \text{if } 5 \leq x \leq 25 \\ 1, & \text{if } x \geq 25 \end{cases}$$

The universe consists of the individuals Sunny, Moon, Pikoo, Gina, Osho, Chang, Paul, Lalu, Lila, and Poly whose ages are 15, 20, 10, 27, 32, 12, 18, 24, 3, and 8 years respectively. Find the *normalcy* of the set as well as *Height* ( $F$ ), *Support* ( $F$ ), *Core* ( $F$ ), and *Cardinality* ( $F$ ).

**Solution 2.9** We calculate the membership values of each individual with the help of the membership function to obtain the following fuzzy set

$$F = \frac{0.25}{\text{Sunny}} + \frac{0.5625}{\text{Moon}} + \frac{0.0625}{\text{Pikoo}} + \frac{1.0}{\text{Gina}} + \frac{1.0}{\text{Osho}} + \frac{0.1225}{\text{Chang}} + \frac{0.4225}{\text{Paul}} + \frac{0.9025}{\text{Lalu}} + \frac{0}{\text{Lila}} + \frac{0.0225}{\text{Poly}}$$

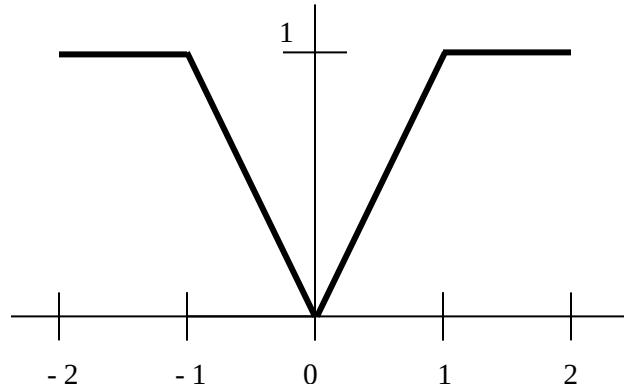
The set is normal because there are two members, Gina and Osho, who attain full memberships. Obviously,  $\text{Height} ( F ) = 1.0$ .  $\text{Support} ( F ) = \{ \text{Sunny}, \text{Moon}, \text{Pikoo}, \text{Gina}, \text{Osho}, \text{Chang}, \text{Paul}, \text{Lalu}, \text{Poly} \}$ ,  $\text{Core} ( F ) = \{ \text{Gina}, \text{Osho} \}$ , and  $\text{Cardinality} ( F ) = 0.25 + 0.5625 + 0.0625 + 1.0 + 1.0 + 0.1225 + 0.4225 + 0.9025 + 0 + 0.0225 = 4.345$ .

♣

**Problem 2.10** Fig. 2.34 shows a membership profile in the form of a reverse triangle. It is defined as follows :

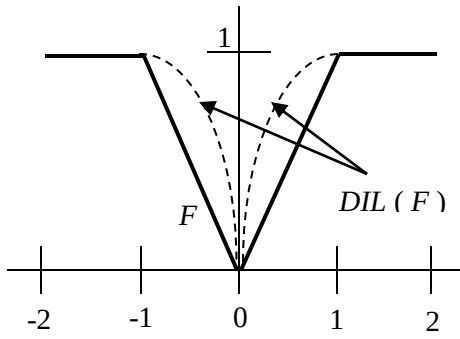
$$\mu_F(x) = \begin{cases} -x, & \text{if } -1 \leq x \leq 0 \\ x, & \text{if } 0 \leq x \leq 1 \\ 1, & \text{elsewhere} \end{cases}$$

Apply the transforms of normalization, dilation, concentration, contrast intensification, and fuzzification on the membership function and show the resultant profiles.

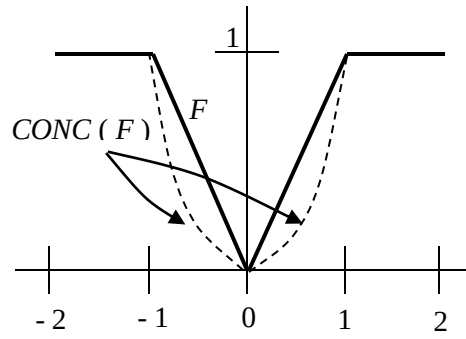


**Fig. 2.34.** A reverse triangular membership function

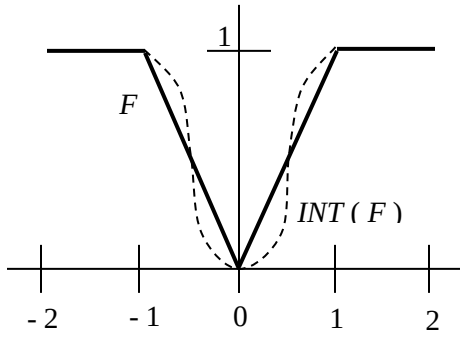
**Solution 2.10** The given membership function is already normal and therefore normalization leaves the function unaltered. To obtain the shape of a transformed membership function, membership values at individual points are calculated. Then these values are mapped to the transformed membership values. For example, let us take the point  $x = -0.5$  so that  $\mu_F(x) = 0.5$ . Under dilation,  $\text{DIL}(\mu_F(x) = 0.5) = (0.5)^{0.5} = 0.71$  (approximate). On the other hand, under concentration, contrast intensification or fuzzification, this membership value maps to  $\text{CON}(\mu_F(x) = 0.5) = (0.5)^2 = 0.25$ ,  $\text{INT}(\mu_F(x) = 0.5) = 2 \times (0.5)^2 = 0.5$ , and  $\text{FUZZ}(\mu_F(x) = 0.5) = (0.5/2)^{0.5} = 0.5$  respectively. The approximate shapes of the transformed functions are shown with dotted line in Fig. 2.35 (a), Fig. 2.35 (b), Fig. 2.35 (c), and Fig. 2.35 (d).



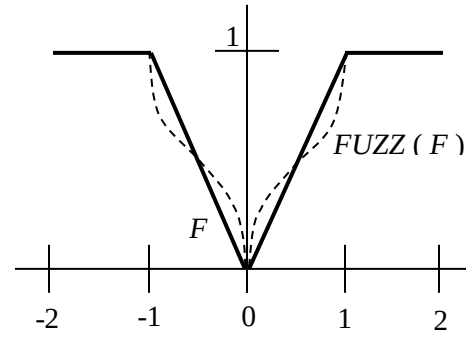
**Fig. 2.35 (a).** Dilation



**Fig. 2.35 (b).** Concentration



**Fig. 2.35 (c).** Contrast intensification



**Fig. 2.35 (d).** Fuzzification

**Fig. 2.35.** Transformations on the reverse triangular function

**Problem 2.11** Let  $U = \text{Flowers} = \{ \text{Jasmine}, \text{Rose}, \text{Lotus}, \text{Daffodil}, \text{Sunflower}, \text{Hibiscus}, \text{Chrysanthemum} \}$  be a universe on which two fuzzy sets, one of *Beautiful flowers* and the other one of *Fragrant flowers* are defined as shown below.

$P = \text{Beautiful flowers} =$

$$\frac{0.3}{\text{Jasmine}} + \frac{0.9}{\text{Rose}} + \frac{1.0}{\text{Lotus}} + \frac{0.7}{\text{Daffodil}} + \frac{0.5}{\text{Sunflower}} + \frac{0.4}{\text{Hibiscus}} + \frac{0.6}{\text{Chrysanthemum}}$$

$Q = \text{Fragrant flowers} =$

$$\frac{1.0}{\text{Jasmine}} + \frac{1.0}{\text{Rose}} + \frac{0.5}{\text{Lotus}} + \frac{0.2}{\text{Daffodil}} + \frac{0.2}{\text{Sunflower}} + \frac{0.1}{\text{Hibiscus}} + \frac{0.4}{\text{Chrysanthemum}}$$

Compute the fuzzy sets  $P \cup Q$ ,  $P \cap Q$ ,  $P'$ ,  $Q'$ ,  $P - Q$ ,  $P \oplus Q$ . Also, verify that  $P \cup P' \neq U$ ,  $P \cap P' \neq \phi$ .

**Solution 2.11** The results are obtained directly from the definitions of the respective operations.

$$P \cup Q = \frac{1.0}{Jasmin e} + \frac{1.0}{Rose} + \frac{1.0}{Lotus} + \frac{0.7}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.4}{Hibiscus} + \frac{0.6}{Chrysanthemum}$$

$$P \cap Q = \frac{0.3}{Jasmin e} + \frac{0.9}{Rose} + \frac{0.5}{Lotus} + \frac{0.2}{Daffodil} + \frac{0.2}{Sunflower} + \frac{0.1}{Hibiscus} + \frac{0.4}{Chrysanthemum}$$

$$\begin{aligned} P' &= \frac{0.7}{Jasmin e} + \frac{0.1}{Rose} + \frac{0}{Lotus} + \frac{0.3}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.6}{Hibiscus} + \frac{0.4}{Chrysanthemum} \\ &= \frac{0.7}{Jasmin e} + \frac{0.1}{Rose} + \frac{0.3}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.6}{Hibiscus} + \frac{0.4}{Chrysanthemum} \end{aligned}$$

$$\begin{aligned} Q' &= \frac{0}{Jasmin e} + \frac{0}{Rose} + \frac{0.5}{Lotus} + \frac{0.8}{Daffodil} + \frac{0.8}{Sunflower} + \frac{0.9}{Hibiscus} + \frac{0.6}{Chrysanthemum} \\ &= \frac{0.5}{Lotus} + \frac{0.8}{Daffodil} + \frac{0.8}{Sunflower} + \frac{0.9}{Hibiscus} + \frac{0.6}{Chrysanthemum} \end{aligned}$$

$$\begin{aligned} P - Q &= P \cap Q' \\ &= \frac{0}{Jasmin e} + \frac{0}{Rose} + \frac{0.5}{Lotus} + \frac{0.7}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.4}{Hibiscus} + \frac{0.6}{Chrysanthemum} \end{aligned}$$

$$\begin{aligned} P \oplus Q &= (P \cap Q') \cup (P' \cap Q) \\ &= \left( \frac{0}{Jasmin e} + \frac{0}{Rose} + \frac{0.5}{Lotus} + \frac{0.7}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.4}{Hibiscus} + \frac{0.6}{Chrysanthemum} \right) \cup \\ &\quad \left( \frac{0.7}{Jasmin e} + \frac{0.1}{Rose} + \frac{0}{Lotus} + \frac{0.2}{Daffodil} + \frac{0.2}{Sunflower} + \frac{0.1}{Hibiscus} + \frac{0.4}{Chrysanthemum} \right) \\ &= \frac{0.7}{Jasmin e} + \frac{0.1}{Rose} + \frac{0.5}{Lotus} + \frac{0.7}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.4}{Hibiscus} + \frac{0.6}{Chrysanthemum} \end{aligned}$$

$$\begin{aligned} P \cup P' &= \frac{0.7}{Jasmin e} + \frac{0.9}{Rose} + \frac{1.0}{Lotus} + \frac{0.7}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.6}{Hibiscus} + \frac{0.6}{Chrysanthemum} \neq U \end{aligned}$$

$$\begin{aligned} P \cap P' &= \frac{0.3}{Jasmin e} + \frac{0.1}{Rose} + \frac{0}{Lotus} + \frac{0.3}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.4}{Hibiscus} + \frac{0.4}{Chrysanthemum} \neq \phi \end{aligned}$$



**Problem 2.12** Let  $A = \{ \text{Gogo, Didi, Pozzo, Lucky} \}$  be a set of individuals and  $B = \{ \text{Colour, Game, Flower, Pet} \}$ . TABLE 2.10 presents the liking data of these individuals with respect to the listed aspects. We define a relation  $R$  on  $A \times A$  such that  $x R y$  if there is which both  $x$  and  $y$  likes. Show the relation matrix for  $R$ .

**TABLE 2.10.** LIKING DATA OF THESE INDIVIDUALS

|              | Likes        |                 |              |               |
|--------------|--------------|-----------------|--------------|---------------|
|              | Colour       | Game            | Flower       | Pet           |
| <b>Gogo</b>  | <i>Blue</i>  | <i>Tennis</i>   | <i>Rose</i>  | <i>Parrot</i> |
| <b>Didi</b>  | <i>Red</i>   | <i>Baseball</i> | <i>Lotus</i> | <i>Parrot</i> |
| <b>Pozzo</b> | <i>Green</i> | <i>Soccer</i>   | <i>Lotus</i> | <i>Dog</i>    |
| <b>Lucky</b> | <i>Red</i>   | <i>Tennis</i>   | <i>Rose</i>  | <i>Cat</i>    |

**Solution 2.12** The relation matrix is given in TABLE 2.10.

**TABLE 2.11.** RELATION MATRIX FOR  $R$

|              | $R$  |      |       |       |
|--------------|------|------|-------|-------|
|              | Gogo | Didi | Pozzo | Lucky |
| <b>Gogo</b>  | 1    | 1    | 0     | 1     |
| <b>Didi</b>  | 1    | 1    | 1     | 1     |
| <b>Pozzo</b> | 0    | 1    | 1     | 0     |
| <b>Lucky</b> | 1    | 1    | 0     | 1     |

**Problem 2.13** Along with relation  $R$  of Problem 2.12, consider one more relation among Gogo, Didi, Pozzo, Lucky regarding who is fond of whom, as shown in TABLE 2.11. We define a relation  $S$  such that  $x S y$  if there exists  $z$  for which  $x$  is fond of  $z$  and  $z$  is fond of  $y$ . Show the relation matrix of  $S$  and then find  $R \cup S$ ,  $R \cap S$ ,  $R \circ S$ .

**TABLE 2.12.** WHO IS FOND OF WHOM

|              | is fond of |      |       |       |
|--------------|------------|------|-------|-------|
|              | Gogo       | Didi | Pozzo | Lucky |
| <b>Gogo</b>  | 1          | 0    | 1     | 1     |
| <b>Didi</b>  | 0          | 0    | 0     | 1     |
| <b>Pozzo</b> | 0          | 0    | 1     | 0     |
| <b>Lucky</b> | 1          | 1    | 0     | 0     |

**Solution 2.13** The relation matrix of  $S$  is shown in TABLE 2.13. TABLE 2.14 and TABLE 2.15 present the relation matrices for  $R \cup S$  and  $R \cap S$ .

TABLE 2.13. RELATION MATRIX FOR  $S$ 

|              | $S$  |      |       |       |
|--------------|------|------|-------|-------|
|              | Gogo | Didi | Pozzo | Lucky |
| <b>Gogo</b>  | 1    | 1    | 1     | 1     |
| <b>Didi</b>  | 1    | 1    | 0     | 0     |
| <b>Pozzo</b> | 0    | 0    | 1     | 0     |
| <b>Lucky</b> | 1    | 0    | 1     | 1     |

TABLE 2.14. RELATION MATRIX FOR  $R \cup S$ 

|              | $R \cup S$ |      |       |       |
|--------------|------------|------|-------|-------|
|              | Gogo       | Didi | Pozzo | Lucky |
| <b>Gogo</b>  | 1          | 1    | 1     | 1     |
| <b>Didi</b>  | 1          | 1    | 1     | 1     |
| <b>Pozzo</b> | 0          | 1    | 1     | 0     |
| <b>Lucky</b> | 1          | 1    | 1     | 1     |

TABLE 2.15. RELATION MATRIX FOR  $R \cap S$ 

|              | $R \cap S$ |      |       |       |
|--------------|------------|------|-------|-------|
|              | Gogo       | Didi | Pozzo | Lucky |
| <b>Gogo</b>  | 1          | 1    | 0     | 1     |
| <b>Didi</b>  | 1          | 1    | 0     | 0     |
| <b>Pozzo</b> | 0          | 0    | 1     | 0     |
| <b>Lucky</b> | 1          | 0    | 0     | 1     |

Now let us compute  $R \circ S$ . Does  $(\text{Gogo}, \text{Gogo}) \in R \circ S$ ? We see,  $(\text{Gogo}, \text{Gogo}) \in R$  and  $(\text{Gogo}, \text{Gogo}) \in S$ . Moreover,  $(\text{Gogo}, \text{Didi}) \in R$  and  $(\text{Didi}, \text{Gogo}) \in S$ . Hence  $(\text{Gogo}, \text{Gogo}) \in R \circ S$ . However there is no  $x$  such that  $(\text{Pozzo}, x) \in R$  and  $(x, \text{Lucky}) \in S$ . Other members of  $R \circ S$  are found in similar fashion ( see TABLE 2.16 ).

TABLE 2.16. RELATION MATRIX FOR  $R \circ S$ 

|              | $R \cap S$ |      |       |       |
|--------------|------------|------|-------|-------|
|              | Gogo       | Didi | Pozzo | Lucky |
| <b>Gogo</b>  | 1          | 1    | 0     | 1     |
| <b>Didi</b>  | 1          | 1    | 0     | 0     |
| <b>Pozzo</b> | 0          | 0    | 1     | 0     |
| <b>Lucky</b> | 1          | 0    | 0     | 1     |

**Problem 2.14** Show that Cartesian product of crisp sets is a special case of fuzzy Cartesian product.

**Solution 2.14** Let  $A$  and  $B$  be two fuzzy sets on reference sets  $X$  and  $Y$  respectively. As per definition of fuzzy Cartesian product  $\mu_{A \times B}(a, b) = \min \{ \mu_A(a), \mu_B(b) \} \forall a \in X, \forall b \in Y$ . Now, if  $A$  and  $B$  are crisp sets, then  $\mu_A(a) = \mu_B(b) = 1, \forall a \in X, \forall b \in Y$ . Therefore,  $\mu_{A \times B}(a, b) = \min \{ \mu_A(a), \mu_B(b) \} = 1$ . Hence Cartesian product of crisp sets is a special case of fuzzy Cartesian product. ♣

**Problem 2.15**  $A = \{ \text{Jack, Lucy, Harry} \}, B = \{ \text{Flute, Drum, Violin, Piano} \}, C = \{ \text{String, Wind, Percussion} \}$  on which the relation  $R = \text{Plays} \subseteq P \times Q$  and  $S = \text{instrument type} \subseteq Q \times R$  is defined ( TABLE 2.17 and 2.18 ). Find  $R \circ S$ .

**TABLE 2.17. RELATION MATRIX FOR  $R$**

|       | $R$   |      |        |       |
|-------|-------|------|--------|-------|
|       | Flute | Drum | Violin | Piano |
| Jack  | 0     | 1    | 0      | 1     |
| Lucy  | 1     | 0    | 1      | 1     |
| Harry | 1     | 1    | 1      | 0     |

**TABLE 2.18. RELATION MATRIX FOR  $S$**

|        | $S$    |      |            |
|--------|--------|------|------------|
|        | String | Wind | Percussion |
| Flute  | 0      | 1    | 0          |
| Drum   | 0      | 0    | 1          |
| Violin | 1      | 0    | 0          |
| Piano  | 1      | 0    | 0          |

**Solution 2.15** TABLE 2.19 shows the relation matrix for  $R \circ S$ .

**TABLE 2.19. RELATION MATRIX FOR  $R \circ S$**

|       | $S$    |      |            |
|-------|--------|------|------------|
|       | String | Wind | Percussion |
| Jack  | 1      | 0    | 1          |
| Lucy  | 1      | 1    | 0          |
| Harry | 1      | 1    | 1          |

♣

**Problem 2.16** Let  $A = \{ \text{Mimi, Bob, Kitty, Jina} \}$  be a set of four children,  $B = \{ \text{Tintin, Asterix, Phantom, Mickey} \}$  be a set of four comic characters, and  $C = \{ \text{funny, cute, dreamy} \}$  be a set of three attributes. The fuzzy relations  $R = x \text{ Likes } y$  is defined on  $A \times B$  and  $S = x \text{ IS } y$  is defined on  $B \times C$  as shown in TABLES 2.20 and TABLE 2.21. Find  $R \circ S$ .

**TABLES 2.20.** RELATION MATRIX FOR  $R = x \text{ Likes } y$

|              | <b>R</b>      |                |                |               |
|--------------|---------------|----------------|----------------|---------------|
|              | <b>Tintin</b> | <b>Asterix</b> | <b>Phantom</b> | <b>Mickey</b> |
| <b>Mimi</b>  | 0.8           | 0.5            | 0.7            | 0.8           |
| <b>Bob</b>   | 0.4           | 0.9            | 0.3            | 0.3           |
| <b>Kitty</b> | 0.6           | 0.7            | 0.4            | 0.9           |
| <b>Jina</b>  | 0.3           | 0.8            | 0.2            | 0.5           |

**TABLES 2.21.** RELATION MATRIX FOR  $S = x \text{ IS } y$

|                | <b>S</b>     |             |               |
|----------------|--------------|-------------|---------------|
|                | <b>funny</b> | <b>cute</b> | <b>dreamy</b> |
| <b>Tintin</b>  | 0.6          | 0.7         | 0.3           |
| <b>Asterix</b> | 0.8          | 0.4         | 0.2           |
| <b>Phantom</b> | 0.1          | 0.2         | 0.1           |
| <b>Mickey</b>  | 0.9          | 0.8         | 0.3           |

**TABLES 2.22.** RELATION MATRIX FOR  $R \circ S$

|              | <b>S</b>     |             |               |
|--------------|--------------|-------------|---------------|
|              | <b>funny</b> | <b>cute</b> | <b>dreamy</b> |
| <b>Mimi</b>  | 0.8          | 0.8         | 0.3           |
| <b>Bob</b>   | 0.8          | 0.4         | 0.3           |
| <b>Kitty</b> | 0.9          | 0.8         | 0.3           |
| <b>Jina</b>  | 0.8          | 0.5         | 0.3           |

**Solution 2.16** Let us recall the definition of composition of fuzzy relations. If  $A$ ,  $B$ , and  $C$  are three crisp sets and  $R$  and  $S$  are fuzzy relations over  $A \times B$  and  $B \times C$ , then the composition of  $R$  and  $S$ ,  $R \circ S$ , is a relation over  $A \times C$  such that

$$(R \circ S)(x, y) = \max_{y \in B} \{ \min(R(x, y), S(y, z)) \}$$

Now,  $R \circ S ( \text{Mimi, funny} ) = \max \{ \min ( R ( \text{Mimi, Tintin} ), S ( \text{Tintin, funny} ) ), \min ( R ( \text{Mimi, Asterix} ), S ( \text{Asterix, funny} ) ), \min ( R ( \text{Mimi, Phantom} ), S ( \text{Phantom, funny} ) ) \}$



),  $\min ( R ( \text{Mimi}, \text{Mickey} ), S ( \text{Mickey}, \text{funny} ) ) \} = \max \{ \min ( 0.8, 0.6 ), \min ( 0.5, 0.8 ), \min ( 0.7, 0.1 ), \min ( 0.8, 0.9 ) \} = \max \{ 0.6, 0.5, 0.1, 0.8 \} = 0.8$ . The membership values for the rest of the ordered pairs can be computed similarly. The resultant relation matrix is shown in TABLE 2.22. ♣

**Problem 2.17** A fuzzy set  $N$  of naughty children is given below. Find the level set of  $N$ .

$$N = \frac{0.3}{\text{Piku}} + \frac{0.8}{\text{Mini}} + \frac{1.0}{\text{Lotus}} + \frac{0.3}{\text{Jojo}} + \frac{0.5}{\text{Nina}} + \frac{0.5}{\text{Joy}} + \frac{0.8}{\text{Lila}}$$

**Solution 2.17** The level set of  $N$ ,  $L ( N )$ , is the crisp set of real values  $x \in ( 0, 1 ]$  such that for each  $x \in L ( F )$  there is a singleton  $( y, x ) \in F$ . Symbolically,  $L ( N ) = \{ x \mid 0 < x \leq 1 \text{ and } \exists y \in U \text{ such that } \mu_N ( y ) = x \}$ . Hence

$$L ( N ) = \{ 0.3, 0.5, 0.8, 1.0 \}. \quad \clubsuit$$

**Problem 2.18** A fuzzy set  $F$  has the membership profile given below. What is the level set of  $F$  ?

$$\mu_F ( x ) = \frac{1}{1 + e^{-x}}, \quad -\infty \leq x \leq \infty$$

**Solution 2.18** The level set is  $L ( N ) = ( 0, 1 )$ . ♣

**Problem 2.19** For the naughty children of Problem 2.17, find  $N_\alpha$  for  $\alpha = 0.4$  and 0.2.

**Solution 2.19** The  $\alpha$ -cut of  $N$ ,  $N_\alpha$ , is the crisp set containing those elements whose membership value with respect to  $N$  is greater than, or equal to,  $\alpha$ . Symbolically,  $N_\alpha = \{ x \mid \mu_N ( x ) \geq \alpha \}$ . Accordingly we find,

$$\begin{aligned} N_{0.4} &= \{ x \mid \mu_N ( x ) \geq 0.4 \} = \{ \text{Mini}, \text{Lotus}, \text{Nina}, \text{Joy}, \text{Lila} \}, \text{ and} \\ N_{0.2} &= \{ x \mid \mu_N ( x ) \geq 0.2 \} = \{ \text{Piku}, \text{Mini}, \text{Lotus}, \text{Jojo}, \text{Nina}, \text{Joy}, \text{Lila} \} \quad \clubsuit \end{aligned}$$

**Problem 2.20** For  $F$  of Problem 2.18, find  $F_\alpha$  for  $\alpha = 0.5$  and 0.

**Solution 2.20**  $F_{0.5} = [ 0, \infty )$  and  $F_0 = ( -\infty, \infty )$ . ♣

**Problem 2.21** Find the fuzzy cardinality of a fuzzy set  $LC$  of low calorie food. The fuzzy set is given below.

$$LC = \frac{1}{\text{Cucumber}} + \frac{1}{\text{Watermelon}} + \frac{0.5}{\text{Chicken}} + \frac{0.1}{\text{Ice-cream}} + \frac{0.1}{\text{Chocolate}} + \frac{0.3}{\text{Rice}} + \frac{0.8}{\text{Egg}}$$

**Solution 2.21** Let  $F$  be a fuzzy set and  $L(F)$  is the level set of  $F$  and for all  $\alpha \in L(F)$ ,  $F_\alpha$  is the  $\alpha$ -cut of  $F$ . The fuzzy cardinality of  $F$ ,  $f_c(F)$ , is the fuzzy set

$$f_c(F) = \sum_{\alpha \in L(F)} \frac{\alpha}{|F_\alpha|}$$

In the present instance,  $L(LC) = \{0.1, 0.3, 0.5, 0.8, 1.0\}$ . The  $\alpha$ -cuts of  $LC$  for  $\alpha = 0.1, 0.3, 0.5, 0.8$  and  $1.0$  are

$LC_{0.1} = \{\text{Ice-cream, Chocolate, Rice, Chicken, Egg, Cucumber, Watermelon}\}$ , and  $|LC_{0.1}| = 7$ ,  $LC_{0.3} = \{\text{Rice, Chicken, Egg, Cucumber, Watermelon}\}$ , and  $|LC_{0.3}| = 5$ ,  $LC_{0.5} = \{\text{Chicken, Egg, Cucumber, Watermelon}\}$ , and  $|LC_{0.5}| = 4$ ,  $LC_{0.8} = \{\text{Egg, Cucumber, Watermelon}\}$ , and  $|LC_{0.8}| = 3$ ,  $LC_{1.0} = \{\text{Cucumber, Watermelon}\}$ , and  $|LC_{1.0}| = 2$ . Therefore the fuzzy cardinality of  $LC$  is given by

$$f_c(LC) = \sum_{\alpha \in L(F)} \frac{\alpha}{|LC_\alpha|} = \frac{0.1}{7} + \frac{0.3}{5} + \frac{0.5}{4} + \frac{0.8}{3} + \frac{1.0}{2}$$

♣

**Problem 2.22** Consider two sets of colours  $C-1 = \{\text{Red, Blue, Green}\}$  and  $C-2 = \{\text{Red, Blue, Green, Yellow, Cyan, Magenta}\}$ . A function  $f: C-1 \times C-1 \rightarrow C-2$  is defined as shown in the matrix below.

|       | Red     | Blue    | Green  |
|-------|---------|---------|--------|
| Red   | Red     | Magenta | Yellow |
| Blue  | Magenta | Blue    | Cyan   |
| Green | Yellow  | Cyan    | Green  |

We define two shades of the colours in  $\{\text{Red, Blue, Green}\}$  as fuzzy sets  $S-1$  and  $S-2$ .

$$S-1 = \frac{0.3}{\text{red}} + \frac{0.5}{\text{blue}} + \frac{0.7}{\text{green}} \quad S-2 = \frac{0.8}{\text{red}} + \frac{0.2}{\text{blue}} + \frac{0.4}{\text{green}}$$

Extend the function  $f$  to the domain  $S-1 \times S-2$  by applying the fuzzy extension principle.

**Solution 2.22** Let us denote the fuzzy equivalent of  $f$  as  $ff$ . The pre-images of the elements of  $C-2$  under  $f$  are

$$\begin{aligned} f^{-1}(R) &= \{ (R, R) \} \\ f^{-1}(G) &= \{ (G, G) \} \\ f^{-1}(B) &= \{ (B, B) \} \\ f^{-1}(M) &= \{ (R, B), (B, R) \} \\ f^{-1}(Y) &= \{ (R, G), (G, R) \} \\ f^{-1}(C) &= \{ (B, G), (G, B) \} \end{aligned}$$

Computations of the fuzzy memberships of the elements of  $C-2$  under  $ff$  are shown below.

$$\begin{aligned} ff^{-1}(R) &= \{ (R, R) \} \\ \therefore \mu_{C-2}(R) &= \max \{ \min ( \mu_{s-1}(R), \mu_{s-2}(R) ) \} \\ &= \max \{ \min ( 0.3, 0.8 ) \} \\ &= \max \{ 0.3 \} = 0.3 \end{aligned}$$

$$\begin{aligned} ff^{-1}(B) &= \{ (B, B) \} \\ \therefore \mu_{C-2}(B) &= \max \{ \min ( \mu_{s-1}(B), \mu_{s-2}(B) ) \} \\ &= \max \{ \min ( 0.5, 0.2 ) \} \\ &= \max \{ 0.2 \} = 0.2 \end{aligned}$$

$$\begin{aligned} ff^{-1}(G) &= \{ (G, G) \} \\ \therefore \mu_{C-2}(G) &= \max \{ \min ( \mu_{s-1}(G), \mu_{s-2}(G) ) \} \\ &= \max \{ \min ( 0.7, 0.4 ) \} \\ &= \max \{ 0.4 \} = 0.4 \end{aligned}$$

$$\begin{aligned} ff^{-1}(M) &= \{ (R, B), (B, R) \} \\ \therefore \mu_{C-2}(M) &= \max \{ \min ( \mu_{s-1}(R), \mu_{s-2}(B) ), \min ( \mu_{s-1}(B), \mu_{s-2}(R) ) \} \\ &= \max \{ \min ( 0.3, 0.2 ), \min ( 0.5, 0.8 ) \} \\ &= \max \{ 0.2, 0.5 \} = 0.5 \end{aligned}$$

$$\begin{aligned} ff^{-1}(Y) &= \{ (R, G), (G, R) \} \\ \therefore \mu_{C-2}(Y) &= \max \{ \min ( \mu_{s-1}(R), \mu_{s-2}(G) ), \min ( \mu_{s-1}(G), \mu_{s-2}(R) ) \} \\ &= \max \{ \min ( 0.3, 0.4 ), \min ( 0.7, 0.8 ) \} \\ &= \max \{ 0.3, 0.7 \} = 0.7 \end{aligned}$$

$$\begin{aligned} ff^{-1}(C) &= \{ (B, G), (G, B) \} \\ \therefore \mu_{C-2}(C) &= \max \{ \min ( \mu_{s-1}(B), \mu_{s-2}(G) ), \min ( \mu_{s-1}(G), \mu_{s-2}(B) ) \} \\ &= \max \{ \min ( 0.5, 0.4 ), \min ( 0.7, 0.2 ) \} \\ &= \max \{ 0.4, 0.2 \} = 0.4 \end{aligned}$$

♣

## Test Your Knowledge

- 2.1 Which of the following phenomena is modeled by fuzzy set theory?
- |                |                          |                      |                          |
|----------------|--------------------------|----------------------|--------------------------|
| a) Randomness  | <input type="checkbox"/> | b) Vagueness         | <input type="checkbox"/> |
| c) Uncertainty | <input type="checkbox"/> | d) None of the above | <input type="checkbox"/> |
- 2.2 Which of the following relations hold good for fuzzy sets?
- |                          |                          |                           |                          |
|--------------------------|--------------------------|---------------------------|--------------------------|
| a) $\mu(x) \in [0, 1]$   | <input type="checkbox"/> | b) $\mu(x) \notin [0, 1]$ | <input type="checkbox"/> |
| c) $\mu(x) = 0$ , or $1$ | <input type="checkbox"/> | d) None of the above      | <input type="checkbox"/> |
- 2.3 Which of the following relations hold good for crisp sets?
- |                          |                          |                           |                          |
|--------------------------|--------------------------|---------------------------|--------------------------|
| a) $\mu(x) \in [0, 1]$   | <input type="checkbox"/> | b) $\mu(x) \notin [0, 1]$ | <input type="checkbox"/> |
| c) $\mu(x) = 0$ , or $1$ | <input type="checkbox"/> | d) None of the above      | <input type="checkbox"/> |
- 2.4 Let  $U = \{a, b, c\}$ , and  $P = \frac{0.5}{a} + \frac{0.5}{c}$  be a fuzzy set on  $U$ . Then which of the following is true?
- |                     |                          |                      |                          |
|---------------------|--------------------------|----------------------|--------------------------|
| a) $P$ is normal    | <input type="checkbox"/> | b) $P$ is sub-normal | <input type="checkbox"/> |
| c) Both (a) and (b) | <input type="checkbox"/> | d) None of the above | <input type="checkbox"/> |
- 2.5 Let  $U = \{a, b, c\}$ , and  $P = \frac{0.5}{a} + \frac{0.75}{c}$  be a fuzzy set on  $U$ . Then what is the *height* of  $P$ ?
- |         |                          |         |                          |
|---------|--------------------------|---------|--------------------------|
| a) 0.5  | <input type="checkbox"/> | b) 0.75 | <input type="checkbox"/> |
| c) 1.25 | <input type="checkbox"/> | d) 1.0  | <input type="checkbox"/> |
- 2.6 What is the *support* of the fuzzy set  $P$  cited in item 2.5?
- |                  |                          |            |                          |
|------------------|--------------------------|------------|--------------------------|
| a) $\{a, c\}$    | <input type="checkbox"/> | b) $\{b\}$ | <input type="checkbox"/> |
| c) $\{a, b, c\}$ | <input type="checkbox"/> | d) $\Phi$  | <input type="checkbox"/> |
- 2.7 What is the *core* of the fuzzy set  $P$  cited in item 2.5?
- |                  |                          |            |                          |
|------------------|--------------------------|------------|--------------------------|
| a) $\{a, c\}$    | <input type="checkbox"/> | b) $\{b\}$ | <input type="checkbox"/> |
| c) $\{a, b, c\}$ | <input type="checkbox"/> | d) $\Phi$  | <input type="checkbox"/> |

- 2.8** What is the *cardinality* of the fuzzy set  $P$  cited in item 2.4?
- a) 0.0 ☐ ☐      b) 0.5 ☐ ☐  
c) 1.0 ☐ ☐      d) None of the above ☐ ☐
- 2.9** Which of the following membership functions must have a *height* of 1.0?
- a) Triangular function ☐ ☐      b) Trapezoidal function ☐ ☐  
c) Gaussian function ☐ ☐      d) None of the above ☐ ☐
- 2.10** Which of the following transformations on membership functions of fuzzy sets *enhances* the membership values?
- a) Dilation ☐ ☐      b) Concentration ☐ ☐  
c) Contrast intensification ☐ ☐      d) Fuzzification ☐ ☐
- 2.11** Which of the following transformations on membership functions of fuzzy sets *reduces* the membership values?
- a) Dilation ☐ ☐      b) Concentration ☐ ☐  
c) Contrast intensification ☐ ☐      d) Fuzzification ☐ ☐
- 2.12** Which of the following transformations on membership functions of fuzzy sets *reduces* as well as *enhances* the membership values selectively?
- a) Contrast intensification ☐ ☐      b) Fuzzification ☐ ☐  
c) Both (a) and (b) ☐ ☐      d) None of the above ☐ ☐
- 2.13** Which of the following properties is not satisfied by fuzzy sets  $P, Q, R$ ?
- a)  $(P \cup Q) \cup R = P \cup (Q \cup R)$  ☐ ☐  
b)  $(P \cup Q)' = P' \cap Q'$  ☐ ☐  
c)  $(P')' = P$  ☐ ☐  
d)  $P \cup P' = U$  ☐ ☐
- 2.14** Which of the following properties is not satisfied by arbitrary fuzzy set  $P$ ?
- a)  $P \cup P' = U$  ☐ ☐      b)  $P \cap P' = \Phi$  ☐ ☐  
c) Both (a) and (b) ☐ ☐      d) None of the above ☐ ☐
- 2.15** Which of the following properties is true for arbitrary fuzzy set  $P$ ?
- a)  $P \cup P' \neq U$  ☐ ☐      b)  $P \cap P' \neq \Phi$  ☐ ☐  
c) Both (a) and (b) ☐ ☐      d) None of the above ☐ ☐

**2.16** Which of the following properties does not hold good for Cartesian product of sets?

- |                   |                          |                      |                          |
|-------------------|--------------------------|----------------------|--------------------------|
| a) Commutativity  | <input type="checkbox"/> | b) Associativity     | <input type="checkbox"/> |
| c) Both (a) & (b) | <input type="checkbox"/> | d) None of the above | <input type="checkbox"/> |

**2.17** Let  $F = \frac{0}{0} + \frac{1}{1}$  be a fuzzy set on the universe  $U = \{ 0, 1 \}$ . Which of the following is  $L ( F )$ , i.e., the level set of  $F$ ?

- |                           |                          |                        |                          |
|---------------------------|--------------------------|------------------------|--------------------------|
| a) $L ( F ) = \{ 0 \}$    | <input type="checkbox"/> | b) $L ( F ) = \{ 1 \}$ | <input type="checkbox"/> |
| c) $L ( F ) = \{ 0, 1 \}$ | <input type="checkbox"/> | d) None of the above   | <input type="checkbox"/> |

**2.18** Which of the following is true regarding variation of the size of  $C$  of a fuzzy set as  $\alpha$  increases from 0 to 1?

- |                                        |                          |                                        |                          |
|----------------------------------------|--------------------------|----------------------------------------|--------------------------|
| a) Size of the $\alpha$ -cut increases | <input type="checkbox"/> | b) Size of the $\alpha$ -cut decreases | <input type="checkbox"/> |
| c) They are not related                | <input type="checkbox"/> | d) None of the above                   | <input type="checkbox"/> |

**2.19** Let  $F_\alpha, F_\beta$  be the  $\alpha$ -cuts of a fuzzy set  $F$  such that  $0 \leq \alpha \leq \beta \leq 1$ . Then which of the following is true?

- |                                 |                          |                                 |                          |
|---------------------------------|--------------------------|---------------------------------|--------------------------|
| a) $F_\alpha \subseteq F_\beta$ | <input type="checkbox"/> | b) $F_\alpha \supseteq F_\beta$ | <input type="checkbox"/> |
| c) $F_\alpha = F_\beta$         | <input type="checkbox"/> | d) None of the above            | <input type="checkbox"/> |

**2.20** Let  $F = \frac{0.1}{a} + \frac{0.5}{b} + \frac{1.0}{c}$  be a fuzzy set on the universe  $U = \{ a, b, c \}$ .  $F$  can be decomposed into a number of  $\alpha$ -cuts such that  $F = \bigcup_{\alpha} \alpha.F_{\alpha}$ . How many such  $\alpha$ -cut decompositions exist?

- |             |                          |                      |                          |
|-------------|--------------------------|----------------------|--------------------------|
| a) One      | <input type="checkbox"/> | b) Three             | <input type="checkbox"/> |
| c) Infinite | <input type="checkbox"/> | d) None of the above | <input type="checkbox"/> |

**2.21** Which of the following is a *fuzzyfication* process?

- |                                     |                          |
|-------------------------------------|--------------------------|
| a) Restricted scalar multiplication | <input type="checkbox"/> |
| b) $\alpha$ -cut decomposition      | <input type="checkbox"/> |
| c) Both (a) & (b)                   | <input type="checkbox"/> |
| d) None of the above                | <input type="checkbox"/> |

2.22 Let  $\alpha F$  denotes the fuzzy set obtained through restricted scalar multiplication of  $F$  by  $\alpha$ ,  $0 \leq \alpha \leq 1$ . Then given  $0 \leq \alpha \leq \beta \leq 1$ , and two fuzzy sets  $F$  and  $G$ , which of the following is not true?

- a)  $\alpha (F \cup G) = \alpha F \cup \alpha G$  ☐ ☐      b)  $(\alpha\beta) F = \alpha (\beta F)$  ☐ ☐  
 c)  $\alpha F \subseteq F$  ☐ ☐      d) None of the above ☐ ☐

2.23 Let  $F = \frac{0.5}{a} + \frac{0.5}{b}$  be a fuzzy set. Then the fuzzy cardinality of  $F$  is

- a)  $\frac{0.5}{1}$  ☐ ☐      b)  $\frac{0.5}{2}$  ☐ ☐  
 c)  $\frac{1.0}{2}$  ☐ ☐      d) None of the above ☐ ☐

2.24 Let  $(a, b)$  and  $(c, d)$  be the pre-images of an element  $p$  under a function  $f$ . The fuzzy membership values of  $a, b, c$ , and  $d$  in a fuzzy set are 0.5, 0.4, 0.7 and 0.2 respectively. What is the fuzzy membership of  $p$  when  $f$  is extended to its fuzzy domain?

- a) 0.5 ☐ ☐      b) 0.4 ☐ ☐  
 c) 0.7 ☐ ☐      d) 0.2 ☐ ☐

2.25 Which of the following statements is not true about a linguistic variable?

- a) A linguistic variable has linguistic values ☐ ☐  
 b) The values of a linguistic variable are fuzzy sets ☐ ☐  
 c) Both (a) and (b) ☐ ☐  
 d) None of the above ☐ ☐

### Answers

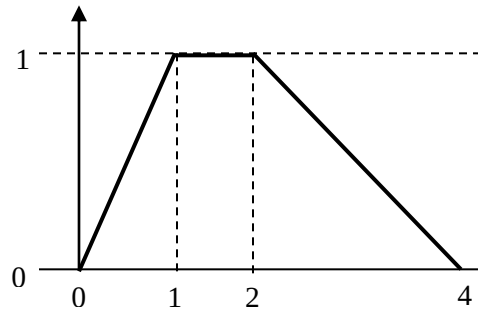
|      |     |      |     |      |     |      |     |      |     |
|------|-----|------|-----|------|-----|------|-----|------|-----|
| 2.1  | (b) | 2.2  | (a) | 2.3  | (c) | 2.4  | (b) | 2.5  | (b) |
| 2.6  | (a) | 2.7  | (d) | 2.8  | (c) | 2.9  | (d) | 2.10 | (a) |
| 2.11 | (b) | 2.12 | (c) | 2.13 | (d) | 2.14 | (c) | 2.15 | (c) |
| 2.16 | (c) | 2.17 | (b) | 2.18 | (b) | 2.19 | (b) | 2.20 | (c) |
| 2.21 | (a) | 2.22 | (d) | 2.23 | (b) | 2.24 | (b) | 2.25 | (c) |

## Exercise

**2.1** Show that set difference operation is not associative, i.e.,  $A - (B - C) \neq (A - B) - C$ .

**2.2** Prove with the help of Venn Diagrams that for arbitrary sets  $A$  and  $B$ ,  $A - B = A \cap B'$

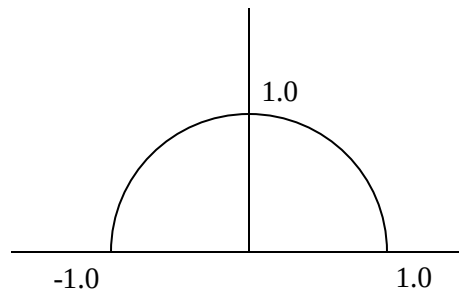
**2.3** Fig. 2.36 shows a trapezoidal membership function with  $a = 0$ ,  $m = 1$ ,  $n = 2$ ,  $b = 4$ . Show the effect of normalization, dilation, concentration, contrast intensification and fuzzification on this membership function.



**Fig. 2.36.** Trapezoidal function  $\mu_T(x, 0, 1, 2, 4)$

**2.4** Fig. 2.37 shows the profile of a membership function defined below. Show the effect of normalization, dilation, concentration, contrast intensification and fuzzification on this membership function.

$$\mu_F(x) = +\sqrt{1-x^2}, \quad -1 \leq x \leq +1$$



**Fig. 2.37.** Semicircular membership profile



**2.5** Let  $P$  and  $Q$  be two fuzzy sets defined on the interval  $[0, \pi]$  with the help of the following membership functions:

$$\mu_P(x) = \sin x, \quad x \in [0, \pi]$$

$$\mu_Q(x) = \begin{cases} \frac{\pi - 2x}{\pi}, & \text{if } 0 \leq x \leq \pi/2 \\ \frac{2x - \pi}{\pi}, & \text{if } \pi/2 \leq x \leq \pi \end{cases}$$

Show the membership function for the fuzzy sets  $P'$ ,  $Q'$ ,  $P \cup Q$ ,  $P \cap Q$ ,  $P - Q$ , and  $P \oplus Q$ .

**2.6** Verify the fuzzy *De Morgan's* law with the fuzzy sets  $A$  and  $B$  cited in Example 2.12.

**2.7** Let  $A = \{0, 1, 2, 3\}$ ,  $B = \{2, 3, 5\}$ , and  $C = \{0, 2, 4\}$  be three sets.  $R$ ,  $S$ , and  $T$  are three relations defined on  $A \times B$ ,  $A \times B$ , and  $B \times C$  respectively. These relations are described below.

$$R = \{ (a, b) \mid (a + b) \text{ is a prime number} \}, R \subseteq A \times B$$

$$S = \{ (a, b) \mid |a - b| \text{ is a prime number} \}, S \subseteq A \times B$$

$$T = \{ (b, c) \mid (b + c) \text{ is a prime number} \}, T \subseteq B \times C$$

Find  $R \cup S$ ,  $R \cap S$ ,  $R'$ ,  $S'$ ,  $T'$ ,  $R \circ T$ , and  $S \circ T$ .

**2.8** Let  $V = \{A, B, C, D\}$  be the set of four kinds of vitamins,  $F = \{f_1, f_2, f_3\}$  be three kinds of fruits containing the said vitamins to various extents, and  $D = \{d_1, d_2, d_3\}$  be the set of three diseases that are caused by deficiency of the vitamins. Vitamin contents of the fruits are expressed with the help of the fuzzy relation  $R$  over  $F \times V$ , and the extent to which the diseases are caused by the deficiency of these vitamins is given by the fuzzy relation  $S$  over  $V \times D$ . Relations  $R$  and  $S$  are given below.

$$R = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} f_1 \\ f_2 \\ f_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.2 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.4 & 0.3 \\ 0.4 & 0.4 & 0.8 & 0.1 \end{bmatrix} \end{matrix}, \quad S = \begin{matrix} & \begin{matrix} d_1 & d_2 & d_3 \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0.3 & 0.5 & 0.7 \\ 0.1 & 0.8 & 0.4 \\ 0.9 & 0.1 & 0.2 \\ 0.5 & 0.5 & 0.3 \end{bmatrix} \end{matrix}$$

Assuming that the max-min composition  $R \circ S$  represents, in some way, the correlation between the amount of a certain fruit that should be taken while suffering from a disease, find  $R \circ S$ .

**2.9** According to the  $\alpha$ -cut decomposition theorem, any fuzzy set  $F$  can be decomposed into a number of  $\alpha$ -cuts,  $F_\alpha$ , such that  $F = \bigcup_{\alpha} \alpha.F_\alpha$ . Usually the  $\alpha$ s are taken from the level set  $L(F)$  of  $F$ . Prove that there exists sets of  $\alpha$ s, other than  $L(F)$ , which satisfies the  $\alpha$ -cut decomposition theorem.

**2.10** Let  $F = \frac{0.6}{a} + \frac{0.2}{b} + \frac{0.3}{c} + \frac{0.9}{d}$  be a fuzzy set. Find a set of  $\alpha$ -cuts such that  $F = \bigcup_{\alpha} \alpha.F_\alpha$ . How many such sets of  $\alpha$ -cuts are there? Justify your answer.

**2.11** The membership function of a fuzzy set *HIG*, where *HIG* stands for *high income group*, is defined as follows:

$$\mu_{HIG}(i) = \begin{cases} 0, & \text{if } i \leq 3 \\ \frac{i-3}{3}, & \text{if } 3 \leq i \leq 6 \\ 1, & \text{if } 6 \leq i \end{cases}$$

The income  $i$  is given in lakh of Rupees per annum. Let  $U = \{ Joy, Jeev, Nina, Simi \}$  be a universe of four persons. Yearly income of *Joy*, *Jeev*, *Nina*, and *Simi* are Rs. 5 lakh, Rs. 8 lakh, Rs. 4 lakh, and Rs. 3.5 lakh respectively. Construct a fuzzy set *Rich* on  $U$  where the richness of a person is given by the membership value of his / her income with respect to the fuzzy set *HIG*. Compute the fuzzy cardinality of *Rich*.

**2.12** Let us consider the crisp sets  $P = Q = R = \{ 0, 1, 2 \}$ . A function  $f: P \times Q \rightarrow R$  is defined as  $f(x, y) = |x - y|$ , for all  $x \in P, y \in Q$ . Show the function table for  $f$ .

Now consider the fuzzy sets  $A$  and  $B$  on the reference sets  $P$  and  $Q$  respectively as given below.

$$A = \frac{1}{0} + \frac{0.5}{1} + \frac{0}{2}, \quad B = \frac{0}{0} + \frac{0.5}{1} + \frac{1}{2}$$

Apply the extension principle to obtain the fuzzy equivalent of the function  $f$  with respect to the fuzzy sets  $A$  and  $B$ ,  $f: A \times B \rightarrow C$ , where  $C$  is the image of  $A \times B$ , defined on the universe  $R$ .

**2.13** Let  $P = Q = \{ 0, 1, 2 \}$  and  $R = \{ 0, 0.5, 1, 1.5, 2 \}$  be crisp sets and  $f: P \times Q \rightarrow R$  be a function signifying the mean of two numbers,  $f(x, y) = (x + y) / 2$ , for all  $x \in P, y \in Q$ . Construct the function table for  $f$ .

Now consider the fuzzy set  $A$  representing the closeness of a number  $x$  to 1 and  $B$  representing its distance from 1.  $A$  and  $B$  are defined on the reference sets  $P$  and  $Q$  respectively. The sets  $A$  and  $B$  may look like

$$A = \frac{0.5}{0} + \frac{1}{1} + \frac{0.5}{2}, \quad B = \frac{1}{0} + \frac{0}{1} + \frac{1}{2}$$

Extend the function  $f: P \times Q \rightarrow R$  to  $f: A \times B \rightarrow C$  with the help of the extension principle where  $C$  is the image of  $A \times B$ , defined on the universe  $R$ .